



The influence of regions of interest on tractography virtual dissection protocols: general principles to learn and to follow

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Abstract

Efficient communication across fields of research is challenging, especially when they are at opposite ends of the physical and digital spectrum. Neuroanatomy and neuroimaging may seem close to each other. When neuroimaging studies try to isolate structures of interest, according to a specific anatomical definition, a variety of challenges emerge. It is a non-trivial task to convert the neuroanatomical knowledge to instructions and rules to be executed in neuroimaging software. In the process called “virtual dissection” used to isolate coherent white matter structure in tractography, each white matter pathway has its own set of landmarks (regions of interest) used as inclusion and exclusion criteria. The ability to segment and study these pathways is critical for scientific progress, yet, variability may depend on region placement, and be influenced by the person positioning the region (i.e., a rater). When raters’ variability is taken into account, the impact made by each region of interest becomes even more difficult to interpret. A delicate balance between anatomical validity, impact on the virtual dissection and raters’ reproducibility emerge. In this work, we investigate this balance by leveraging manual delineation data of a group of raters from a previous study to quantify which set of landmarks and criteria contribute most to variability in virtual dissection. To supplement our analysis, the variability of each pathway with a region-by-region exploration was performed. We present a detailed exploration and description of each region, the causes of variability and its impacts. Finally, we provide a brief overview of the lessons learned from our previous virtual dissection projects and propose recommendations for future virtual dissection protocols as well as perspectives to reach better community agreement when it comes to anatomical definitions of white matter pathways.

Keywords Diffusion MRI · Reproducibility · Segmentation · Tractography · Virtual dissection · Region of interest

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Introduction

Describing the complex architecture of the white matter of the human brain is a challenging task. There is a balance between neuroanatomical accuracy, simplicity and conciseness when one attempts to communicate the knowledge about the cortical terminations, the landmarks surrounding pathways or their shapes (Benedictis et al. 2016; David et al. 2019; Dick and Tremblay 2012; Fernández-Miranda et al. 2015; Hau et al. 2017; Maffei et al. 2018; Vavassori et al. 2021). Diffusion MRI tractography facilitates the process of designing more efficient ways to convey neuroanatomical descriptions at a large scale. Virtual reconstruction of major WM pathways with tractography is a useful tool to probe structural connectivity and has the potential to be useful for teaching neuroanatomy.

Virtual dissection allows the isolation of known pathways of interest by targeting streamlines using inclusion and exclusion regions of interest (ROIs). The instructions to delineate a specific pathway can be detailed and shared using a task protocol. However, tractography reconstructions do not always accurately represent the underlying anatomy (Jbabdi and Johansen-Berg 2011; Maier-Hein et al. 2017; O'Donnell and Pasternak 2015; Rheault et al. 2020b; Schilling et al. 2020). It is important to consider that the algorithmic limitations of tractography will impact virtual dissections in various ways. If inclusion ROIs are too large or exclusion ROIs too small, bundles can be riddled with spurious streamlines (artificially inflating the volume). If inclusion ROIs are too small or exclusion ROIs too large, bundles can appear sparse or disappear entirely (giving the appearance of a damaged/obliterated pathway). If the virtual dissection is not anatomically informed or if the ROIs are not well positioned, the observed bundle can be a false positive in its entirety.

These challenges to virtual dissection can be addressed by creating protocols that detail the segmentation tasks and target specific situations and audiences. For example, a protocol designed for neurosurgical exploration executed by experts in neuroanatomy can be very different from an application in aging with thousands of subjects performed by data scientists. There is nothing that prevents virtual dissection protocols from requiring preprocessing [e.g., Freesurfer (Dale et al. 1999)] to better handle specific situations. However, depending on the intended use or targeted audience it may be ill-advised to rely on external software to provide ROIs for the virtual dissection process. For example, limitations due to pathologies or injuries where the external software fails (e.g., tumors, strokes, etc.) can limit its scope. Limitations due to the technical knowledge or processing capacity of the target audience can also limit the virtual dissection protocol's scope. It is important to

explicitly mention that protocols cannot be (easily) generalized to a new set of variables. For example, transitioning to a different interactive segmentation software or a protocol designed for tractograms reconstructed using diffusion tensor imaging (DTI) is not necessarily valid when applied to a probabilistic tractogram obtained from constrained spherical deconvolution (CSD).

Virtual dissection protocols must take into account the scenarios for which they are designed and the previously mentioned biases to limit wrongful results and erroneous interpretations. There is a delicate balance between maintaining anatomical accuracy and avoiding spurious streamlines (Schilling et al. 2020; Thomas et al. 2014). To avoid inaccuracies (i.e., anatomically implausible streamlines), virtual dissection protocols must target known pathways with unambiguous instructions. Major challenges of protocol design include: clearly describing ROIs position and shape, including only useful/relevant ROIs with predictable effects and anticipating the “spectrum” of spatial effects on bundles caused by ROIs (and if they all are anatomically plausible). It is important to note that while virtual dissection can apply to exploratory research related to WM connectivity, virtual dissection protocols must rely on anatomical priors obtained from other validation methods and require a higher level of confirmation and anatomical ‘certainty’. Virtual dissections and connectomics are two separate approaches to investigate the connectivity of the brain; this work only applies to the former.

In this work, we investigate (1) the variability in region delineation, (2) the importance of ROIs and ROIs placement on resulting bundles and finally (3) we visualize the spatial impact caused by ROIs' variability. We made observations that allow us to extract recommendations that could help to achieve “clear instructions” and improve future protocols to make them more robust. With the careful examination of each ROI intrinsic variability and the effect it has on pathways, recommendations to increase anatomical accuracy, reduce variability and decrease workload for raters were derived. Furthermore, the insight obtained from our investigation can be consolidated to design better ways to *teach* neuroanatomy from tractography (Arantes et al. 2018; Chang and Molnár 2015). It is noteworthy that the goal of this work is not to design a perfect fits-all protocol, not to claim it can be generalized to pathology and not to argue that the anatomical definition is better or more valid than others. The aim of this project is first and foremost to gain insight into white matter pathway segmentation using standardized protocols and how to design, share and evaluate them.

Background

From a neuroimaging perspective, *ex vivo* approaches such as Klingler's dissection (Agrawal et al. 2011), histological

staining/tracing (Alturkistani et al. 2016; Saleeba et al. 2019) or more recent methods such as polarized light imaging (PLI) or optical coherence tomography (OCT) (Axe et al. 2016; Wang et al. 2015) are considered *ground truth* to reconstruction methods from diffusion MRI tractography. However, they should be seen more like a *gold standard* with several caveats to be acknowledged. Large-scale patterns can be agreed upon, but each laboratory has its techniques. This limits the ability to reach an agreement on the anatomical definition. Even when knowing that two regions are indeed connected, the exact path is not as clearly defined as some would expect. While these challenges are relevant to neuroimaging, they must be resolved by neuroanatomists. However, we can design virtual dissection protocols based on definitions even if they are not considered a consensus and explore how to convey such information as well as how to effectively execute virtual dissection tasks.

The reproducibility of virtual dissection protocols can be evaluated even without a strong consensus on the anatomical definition (Rheault et al. 2020a; Schilling et al. 2021a). This is a time-consuming process that requires collaborations from multiple individuals, the careful design of a virtual dissection protocol and a rigorous framework for the analysis. Rheault et al. (2020a, b) (Tractostorm 1) proposed guidelines for robust quantification of reproducibility. By assembling a group of non-experts and experts (with and without a formal background in anatomy) and providing them with triplicate datasets (5 HCP subjects $\times$ 3 copies), variability in the virtual dissection of the pyramidal tracts (Chenot et al. 2019) was quantified. It was shown that experts and non-experts had similar intra-rater and inter-rater (within their respective group) reproducibility scores.

Due to the heavy workload and limitations present in the first project, this approach was expanded with Rheault et al. (2022b) (Tractostorm 2). The study design was modified to investigate agreement between raters over time (using 5 HCP subjects $\times$ 4 copies in a specific order), an online course was added to introduce the virtual dissection and the number of bundles targeted by the protocol was increased (to 3). This work revealed that raters' agreement did not significantly change over time, showed a significant (global) increase in reproducibility as a result of the online course and confirmed that each bundle of interest has its own reproducibility scores and trends.

From the work of Rheault et al. (2022b) (Tractostorm 2) manual delineation of WM landmarks (15) from 20 raters on 20 datasets (5 HCP subjects $\times$ 4 copies) were obtained. This effort led to 6000 regions of interest (ROIs) that were not detailed in the analysis presented in Rheault et al. (2022b). The current work presents this previous set of data by investigating the observed intrinsic variability (position, shape) from the manual delineations as well as the impact on each

pathway caused by the regions of interest involved in the pathway's virtual dissection.

Rheault et al. (2022a) first proposed a distinction between a theoretical definition and a practical definition. The term "Theoretical definition" was introduced as "*must refer to conceptual landmarks and regions, convey the general shape, orientation and terminations. They must aim to be distinct enough between bundles while encompassing a variety of practical definitions. Such definitions should refer to anatomical structures of the brain and not be tied to specific contrasts or processing.*" Practical definitions are analogous to virtual dissection protocols and were introduced as "*to a specific execution of a theoretical definition. They can change according to the target audience (e.g., experts vs. nonexperts), time requirements (e.g., quick approximation vs. careful delineation), visualization software (e.g., MI-Brain, Rheault et al. 2016 vs. TrackVis, Wang et al. 2007 vs. DSI-Studio, Yeh 2021), or underlying processing (e.g., DTI vs. HARDI, deterministic vs. probabilistic). This also applies to automatic dissection methods.*"

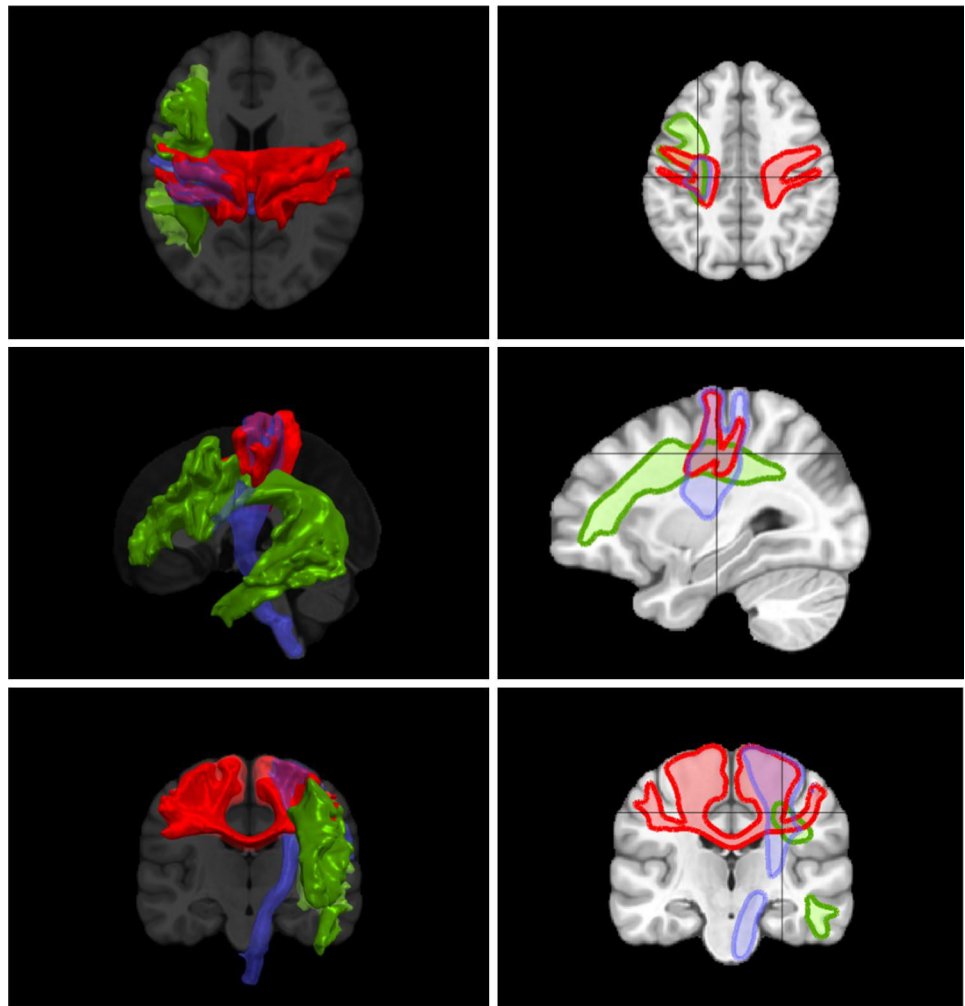
Modification to the processing or execution of a protocol could have small or large effects on variability. This means a new quantification of variability is likely required when changing the previously mentioned variables. The virtual dissection protocol used as a basis for this work is from Rheault et al. (2022b) and the current protocol (Tractostorm 2-Zenodo) is considered a practical definition.

Methods

All data used for this work come from Tractostorm 2 (Rheault et al. 2022b). In summary, the study design involved 20 raters that contributed by carrying out the same pre-established protocol. Each rater had 3 bundle dissections to perform for each of the 20 datasets. The 20 datasets were, in fact, 5 HCP subjects each with 4 duplicates each. In total, 1200 bundles and 6000 regions of interest were submitted.

This specific protocol used by all raters in this study (available at Tractostorm 2-Zenodo) was designed for non-experts in neuroanatomy (basic knowledge of brain organization) with some level of familiarity with tractography. The protocol is adapted for probabilistic tractography from HARDI local models (Garyfallidis et al. 2014; Tournier et al. 2019) and can be carried out in approximately 30 min without external processing apart from the tractography itself. The result of the virtual dissection protocol is a set of three bundles of interest (BOI): one commissural (body of the corpus callosum, CC), one association (left arcuate fasciculus, AF) and one projection (left pyramidal tract, PYT). The expected shape of the pathways can be seen in Fig. 1, the rendering was generated from the *average* BOIs from the submissions (voxel-wise majority vote).

Fig. 1 Average expected shape of bundles segmented using the protocol. The corpus callosum, arcuate fasciculus and pyramidal tract in red, green and blue, respectively. The choice of bundles was made to explore the three main types of pathways (commissural, association and projection) as well as to limit the workload on our collaborators. The isosurfaces were generated at the probability value of 0.5 based on segmentation performed by the principal investigators. The protocol provided to collaborators/raters (available at Tractostorm 2-Zenodo) presented that figure to provide context to the tasks they were about to initiate



A total of 15 regions of interest had to be delineated (some are used for more than one pathway) for the project. There were three main categories of ROIs: Disk region, large hand-drawn plane, whole slice plane. The decision to have only planar regions representing simple shapes was made in an attempt to accelerate the virtual dissection process, facilitate interpretation and keep variability to a minimum. Each ROI had to be anatomically justified and was designed to leave as little as possible to interpretation. Figure 2 shows the 15 ROIs from a random rater on the first HCP dataset. The naming convention for the ROI may seem erroneous at first, but the names were chosen to minimize length and convey their ‘intent’. For example, the postcentral gyrus region does not actually delineate the postcentral gyrus, but is simply a plane attempting to delimit the most posterior section of the postcentral gyrus. Regions such as the midbrain or the medulla oblongata are not delineation of these regions, but ROI to ensure tractography was reaching these as low as these regions (and most importantly continuing to the lower brainstem). To understand the naming convention and the intent of each

ROI, we recommend reading the full protocol description available online.

Analysis

To investigate the intrinsic variability of the positioning and shape of the ROIs, we computed the distance between the center of mass as well as a Hausdorff distance (shape similarity, Huttenlocher et al. 1993) for matching ROIs across raters. These metrics relay information about the disagreement in position (or *difficulty* to position the ROIs) and the disagreement in shape. For the circular ROIs, the disagreement in shape is analogous to variation in the diameter chosen by the rater in the software (Rheault et al. 2016).

To evaluate the impact of ROIs variability on the pathways, we used two approaches: removal of a ROI from a set of dissection criteria and substitution of ROIs between raters.

ROIs used in our virtual dissection protocol are not necessarily all created equal. Some were designed to enforce a known anatomical rule, others to constrain to a specific

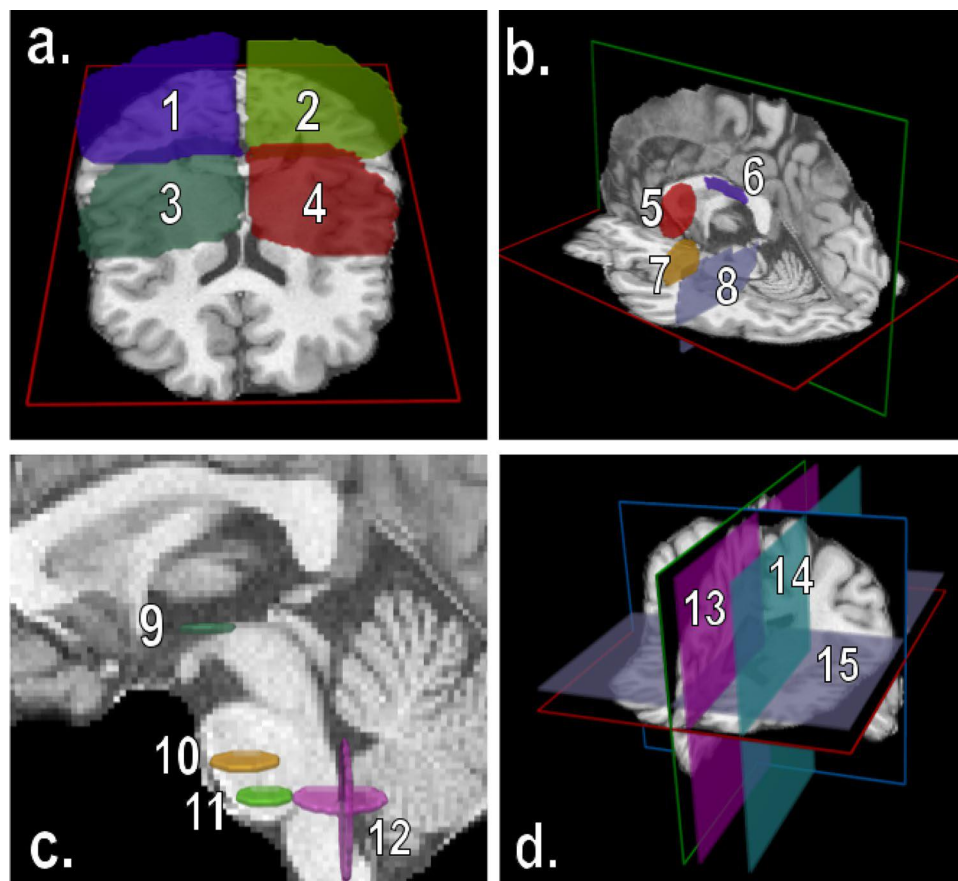


Fig. 2 Rendering of the ROIs from a single HCP subject showing their expected size, shape and position. **(a)** The postcentral (1–2) and precentral (3–4) (left and right), are defined as planes. The planes represent the most anterior region associated with the precentral and the most posterior region associated with the postcentral. These two regions aimed to define the space in between as where the entire pre-central and postcentral gyri are located. **(b)** The centrum semiovale (5), body of the corpus callosum (6), the temporal stem (7) and the temporal lobe entry (8). The regions are defined either as circular-shaped or hand-drawn. **(c)** The internal capsule (9), midbrain (10),

medulla oblongata (11) and cerebellum entry (12). The region's names do not exactly reflect their position, but rather where tractography is expected to reach (or go through) when filtered by the ROI. **(d)** The mid-sagittal plane (13), the medial sagittal limit (14) and the lower axial limit (15) are three planar regions to prevent tractography from reaching a specific area (for more details, Tractostorm 2-Zenodo). Segmentation of the corpus callosum involved ROI #1-2-3-4-6-15, segmentation of the arcuate fasciculus involved ROI #2-4-5-7-8-14, segmentation of the pyramidal tract involved ROI #2-4-9-10-11-12-13

shape to avoid spurious streamlines. However, ROIs interact with each other, leading to redundant criteria and small contributions to the overall virtual dissection. To quantify the impact of an individual ROI on a pathway, we performed virtual dissections and omitted one ROI (at the time) from the set of rules and compared the result to the baseline described above. Each dataset segmented using the complete protocol by the same rater is the associated baseline. This informs us of the overall usefulness of a specific ROI in the virtual dissection protocol.

The substitution approach uses all the ROIs to dissect a BOI to obtain a baseline. Then, the ROI is substituted with one from another rater and is compared to the baseline using the bundle adjacency distance (Garyfallidis et al. 2018; Rheault et al. 2020a). This operation informs us on

the influence a specific ROI's intrinsic variability has on the final result (bundle) and if this influence is the same across the three bundles. The operations are repeated for each ROI, across raters and for each bundle of interest. Since some ROIs are used by more than one bundle, this leads to 19 distributions (CC: 6, AF: 6, PYT: 7) from 15 ROIs.

Finally, an ROI-by-ROI exploration of each BOI's variability caused by each ROI was performed. This material (available in the supplementary materials) provides a detailed description of each region, the causes of variability, its impacts and potential solutions. For each ROI, a figure will accompany the written description. Figure 3 shows the visualization method used to inform how the variability of ROIs translates to the bundles. The color-encoding of streamlines was designed to facilitate

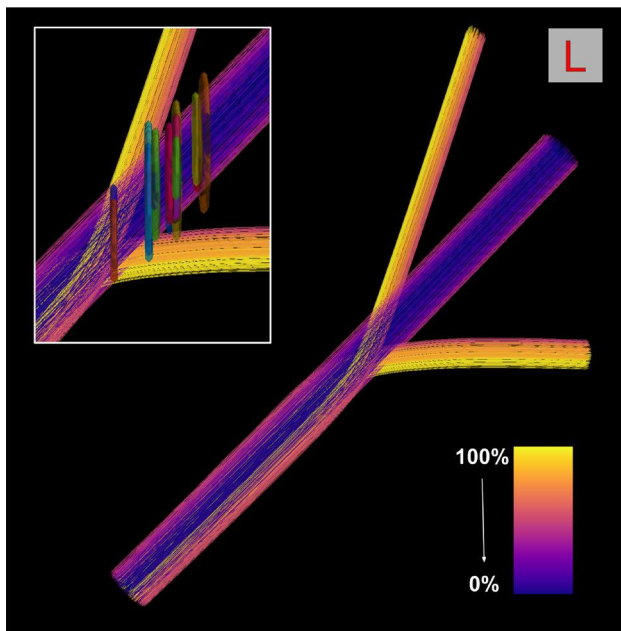


Fig. 3 In this figure, we can observe that the (inclusion) ROIs are generally well positioned in the main axis of the bundle. However, some are placed “too early” and partially intersect the crossing and the effect is that branches are accidentally selected. Variation in shape/size seems to have a minor impact as most of the center bundle is correctly selected. Typically, the more yellow present in the streamlines, the more impactful (and potentially prone to variability) the ROI is in the set of rules to segment the bundles. For example, we can conclude that adding a few millimeters after the crossing as a safe margin for the segmentation would increase reproducibility and guarantee that the center of the bundle would be well defined. The choice to use a subcomponent of the Fibercup (Poupon et al. 2008) is simply to help navigation and illustrate the organization/coloration of streamlines

visualization of the impact of rater variability; this figure shows how an average bundle is impacted by the application of all ROIs. If a streamline is selected/excluded by 1, 2 and up to 80 ROIs this score is mapped to a color spectrum (lower right). Purple means that the streamlines are consistently involved in the segmentation by the raters and yellow means only a few raters selected/excluded the streamlines. The percentage represents the ratio of raters that missed the streamline; 1/80 (purple): almost no rater missed that streamline, 79/80 (yellow): almost all raters missed that streamline).

All summary notes are all organized similarly. The caption of Fig. 3 contains an example of interpretations that are drawn from the visualization. The written text aims to describe:

- The importance and meaning of the anatomical ROI.
- How the raters were expected to locate and draw the ROI.
- What were the sources of confusion and why did it cause so much/little variability?

- What are the ways to reduce variability and is the ROI necessary?
- What are the consequences if a rater misinterpreted the instructions?

This meticulous inspection combined with the previously mentioned analysis is the basis for our summarized recommendation of “Regions of interest impact on pathways” and “Conclusion”. Only an overview of one ROI per bundle is presented in the results section. All 19 summary notes are available in the supplementary materials.

Results

Regions of interest intrinsic variability

Variability of the regions of interest positioning is different for each ROIs. Despite a low average distance, standard deviation indicates that some ROIs are harder to position than others. Regions such as the middle sagittal plane or the ROIs related to the projection pathway (internal capsule, midbrain and medulla) all have an average center of mass distances across raters below 4 mm with a small standard deviation (between 2 and 4 mm) (Fig. 4). However, regions such as the precentral and postcentral gyri or the temporal stem all have a higher average center of mass distances across raters (around 6 mm) and a much larger standard deviation (between 4 and 8 mm). These regions reach translation differences as high as 30 mm. Figure 4 shows the distribution of distances for each ROI and how raters disagree on their placement. The temporal entry region has a small cluster of translation distances above 40 mm, this is due to a single rater accidentally drawing the region on the left and on the right. While this has no influence on the bundle segmentation, it creates a small cluster of 20 points (due to the pairwise comparison of raters). The same can be observed on Fig. 5. It is important to mention that in the virtual dissection protocol (Tractostorm 2-Zenodo), some regions that are not independent (described in relation to others) are therefore subject to similar levels of variability. For example, the centrum semiovale region (central CS) is described as a circular ROI located between the precentral and postcentral gyri.

Disagreement on ROIs’ shape was lower. As a matter of fact, instructions and examples detailing the expected shape constrained the raters to similar delineation. For example, all circular ROIs had a proposed approximate radius in the instructions. This means that the distance shown in Fig. 5 for the internal capsule (IC), midbrain (MB), medulla (MO), centrum semiovale (central CS) and temporal stem were simply the distribution of differences in radius between raters. ROIs covering the entire plane such as the lower

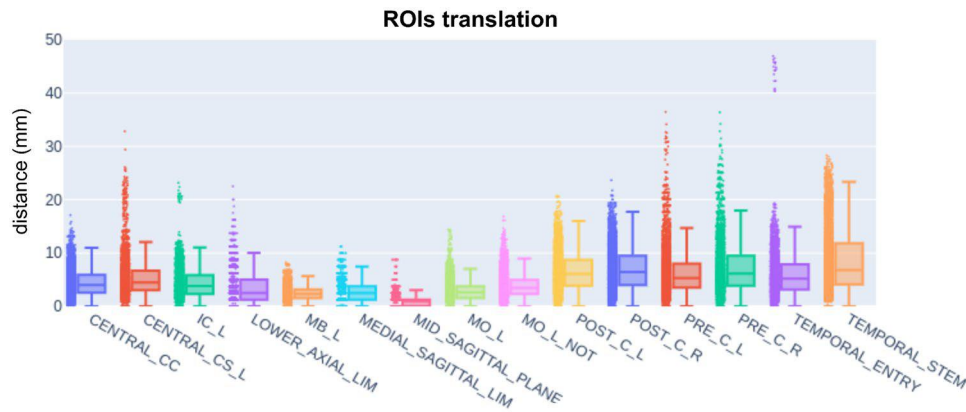


Fig. 4 Translation distances (in mm) between raters segmentation for each ROI (center of mass). Low values mean that raters positioned their region closely (e.g., mid-sagittal plane), high values mean that raters did not agree on positioning and placed their ROI far apart

(e.g., temporal stem). This can be interpreted as the ease (or difficulty) to locate the slice indicated by the protocol. Pairwise comparisons of ROIs were made within HCP datasets; there was no significant inter-subject difference

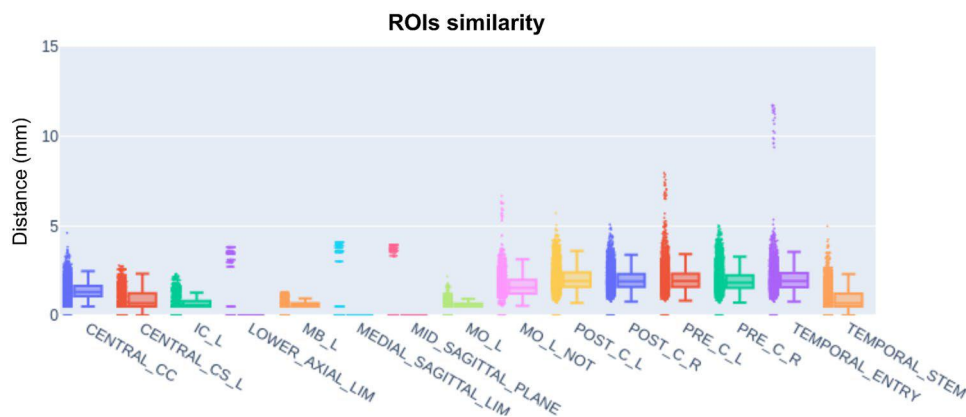


Fig. 5 Similarity distances (in mm) between raters segmentation for each ROI (mean Hausdorff distance). Low values mean that raters drew by hand (or circular region) a similar shape (or radius). High values mean that raters disagree on the shape/size of regions. For example, all three planar regions are extremely similar in shape because most raters selected the entire slice, thus identical shape.

Regions such as the internal capsule or temporal stem are circular and the difference is due to a different choice of radius in the software. This can be interpreted as the ease (or difficulty) to choose the appropriate shape/size of the regions indicated by the protocol. Pairwise comparisons of ROIs were made within HCP datasets; there was no significant inter-subject difference

axial limit, the medial sagittal limit or the middle sagittal plane had little to no shape difference. This is due to the fact that most raters selected whole slices (no shape difference, 0 mm) or drew large squares nearly covering the whole slice (small difference, below 4 mm). The ROIs with the highest variability in shape are the large hand-drawn regions, such as the precentral and postcentral gyri (left and right).

Regions of interest impact on pathways

By removing one ROI from the set of rules during a pathway virtual dissection, we can identify if it is useful to the process, however, this is limited to a single removal as the interaction between ROI could have unexpected consequences on

the final bundle. In Fig. 6, we can see ROIs that are critical to the virtual dissection and others that are not affecting the final bundles. The body of the corpus callosum (central CC) is an example of a crucial ROI, the Dice score between the baseline bundle and the bundle dissected without using the central CC criterion average at 0.25.

Without it, the resulting pathway encompasses commissural, association and projection streamlines and lead to a completely different morphology (closer to a whole-brain than a bundle). The lower axial limit, temporal entry and internal capsule (IC) are examples of important ROIs (Dice scores of 0.8, 0.8 and 0.75, respectively). Removing them changes the resulting bundle, but remains somewhat similar to the baseline. The temporal stem, postcentral gyrus (for

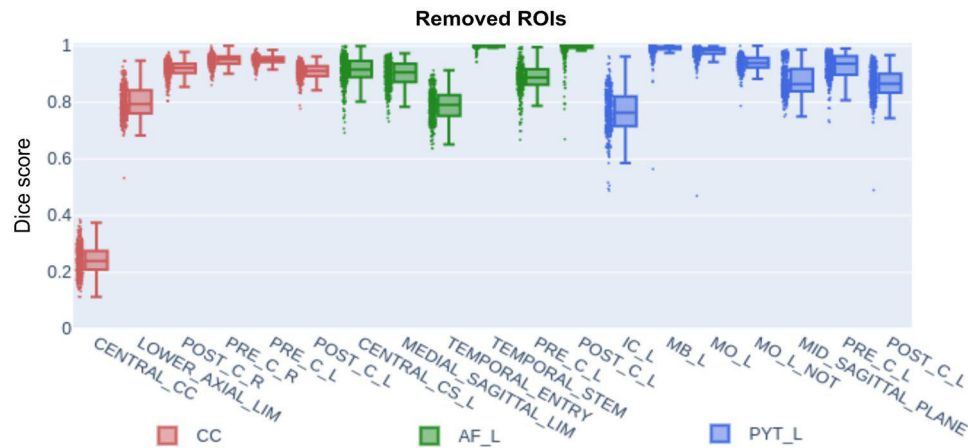


Fig. 6 Dice score representing the overlap of bundles (*RGB*: *CC*, *AF*, *PYT*) when one ROI is removed from the set of segmentation rules. The ROIs shown on the X-axis are the regions removed from the set of rules. High scores with low variability mean that the ROI impact on the final segmentation is negligible (e.g., temporal stem). Low

scores with high variability mean that ROI impact on the final segmentation is substantial (e.g., body of the corpus callosum, *CC*). This can be interpreted as the impact the ROI/rule has on the virtual dissection (is it needed or not in the protocol)

the arcuate fasciculus), and midbrain (*MB*) are examples of ROIs with low impact (all Dice scores averaging at 0.99). Removing these ROIs from the set of rules of the virtual dissection protocol does not significantly affect the shape.

Identifying the ROIs that do not contribute to the virtual dissection protocol is one way to probe the set of rules, but the variability caused by each ROI is another important insight. The intrinsic variability of ROIs does not directly inform us on the resulting bundle variability. Figure 7 shows the variability caused by each ROI when substituted from all raters. ROIs substitution indicates all ROIs contribute (at least partially) to the overall bundle variability. However,

while the average Dice scores are high, some ROIs have a large standard deviation. The central *CC*, temporal entry, precentral gyrus (of the arcuate fasciculus) and the internal capsule (*IC*) have standard deviations above 0.1 with Dice scores lower value reaching as low as 0.3.

The variability caused by a ROI present in each set of rules (three BOIs) is not equal across BOIs, meaning that the interaction with the different rules in the virtual dissection process is likely unique. For example, the precentral gyrus is used as an exclusion criterion in the *CC* and *PYT* and both have very low variability and small standard deviations (meaning that raters' variability for that ROI has a low and

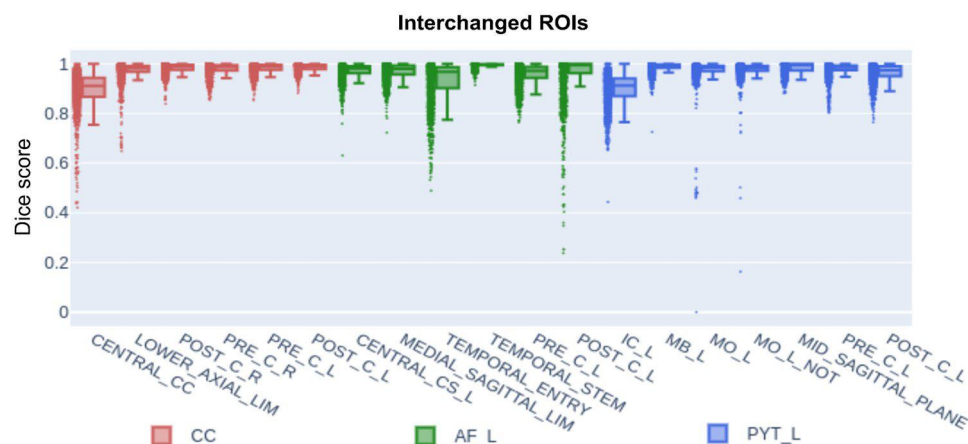


Fig. 7 Dice score representing the overlap of bundles (*RGB*: *CC*, *AF*, *PYT*) when one ROI from a specific rater is substituted with the same ROI from another rater. High scores with low variability mean that the ROI decisions have a limited impact on the final results (e.g., temporal stem, *AF*). Low scores with high variability mean that ROI

decisions will, sometimes, have a major impact on the resulting bundle (e.g., internal capsule, *PYT*). This can be interpreted as how much a ROI variability has on the bundle's virtual dissection variability. The red (*CC*) boxplots are linked to their visual counterpart in Fig. 8. Same for the green (*AF*) with Fig. 11 and blue (*PYT*) with Fig. 9

stable impact on the BOI. However, the same ROI when used as an inclusion criterion for the AF has a much larger standard deviation (meaning that raters' variability is less stable when used with this specific virtual dissection).

This ROI-by-ROI effect is explored in more detail in the supplementary materials. This visualization is necessary because it is difficult to choose a quantitative threshold based on the boxplots. A good Dice Score (e.g., > 0.9) can be achieved while drastically reducing the anatomical validity of a resulting bundle. For this reason, the qualitative inspection of Figs. 8, 9, 10 is crucial to complement the score of Fig. 7. Figures 8, 9, 10 are previews of the visualization technique used to explore how the ROI variability impacts each bundle in the virtual dissection protocol. The yellow coloring represents the lower value of the reproducibility spectrum. In other words, the more the streamlines' color tends towards yellow, the more they are inconsistently

included in the final bundles due to higher variability in the ROI positioning. On the opposite, the more the streamlines' color tends towards dark blue, the more they are included in the bundle. This qualitative visualization provides insight into the spatial impact of each ROI's variability and should be seen as complementary information to Fig. 7.

Discussion

From previous projects, we learned that a protocol can be improved with an online course (Rheault et al. 2022b), we learned that raters are a bigger source of variability than acquisition schemes or data quality (Rheault et al. 2022a), we learned that with appropriate protocol experts and nonexperts have similar levels of reproducibility (Rheault et al. 2020a), we learned that there are ways to quantify

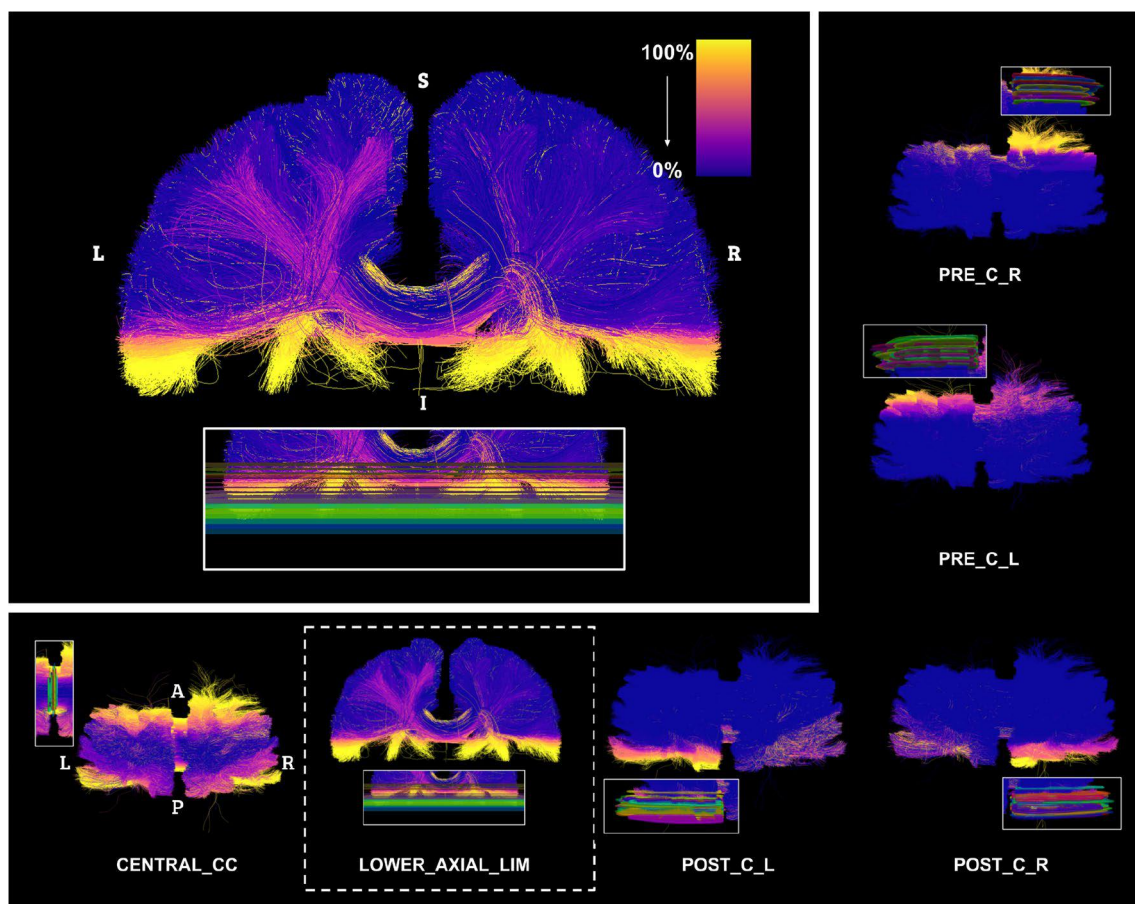


Fig. 8 Mosaic of all ROIs involved in the corpus callosum virtual dissection. Streamlines variability, due to ROI placement, is encoded in the coloring. The lower axial limit is an interesting example of how differences in positioning can affect the bundle. The goal of this region is to prevent streamlines from entering a projection pathway or reaching into the nuclei (exclusion criterion). While the majority of raters selected an axial slice low enough to achieve that goal without

impacting the bundle, a few raters selected an axial plane that started too high and impacted the lateral fanning in the precentral and post-central gyri. The instruction could have been made clearer and the ROI modified to simply be a large rectangle that only blocks the most central region of the brain rather than the entire slice. This way, the same goal is achieved without impacting the fanning regions

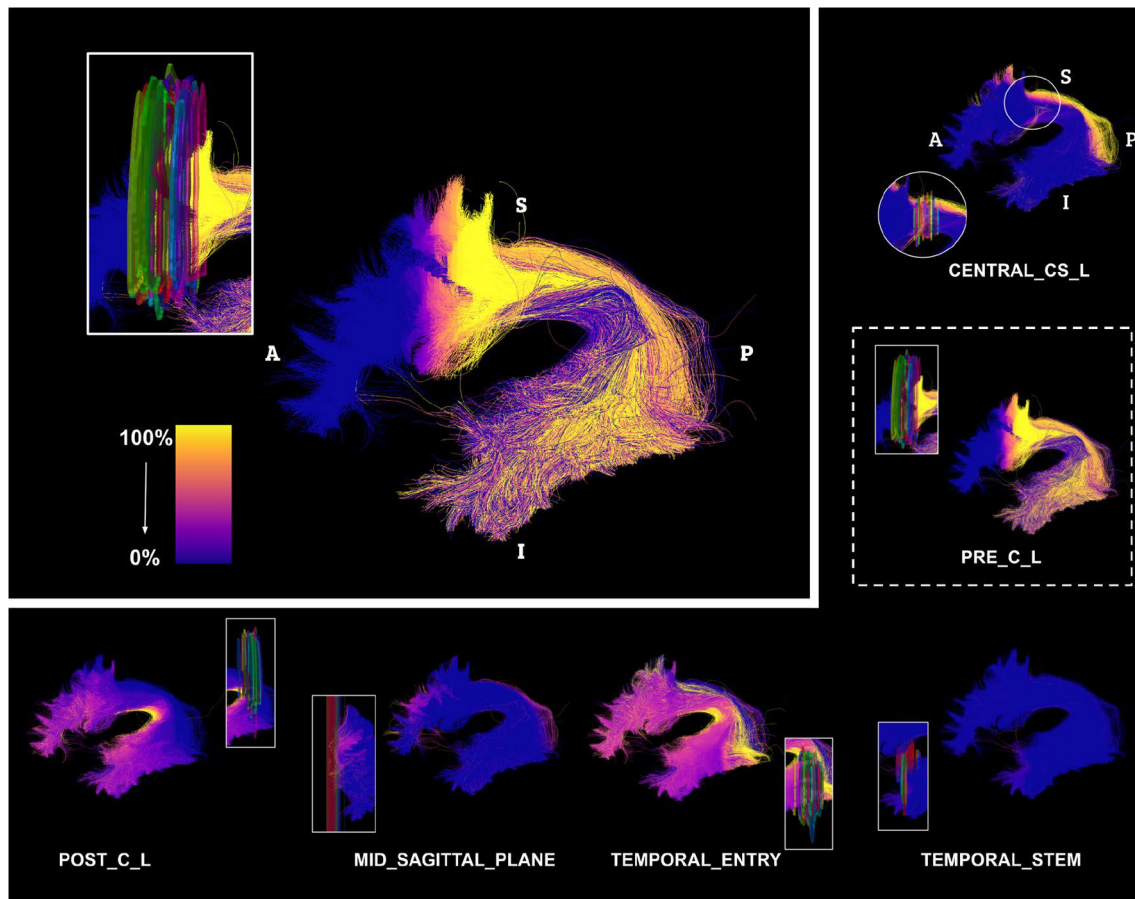


Fig. 9 Mosaic of all ROIs involved in the arcuate fasciculus virtual dissection. Streamlines variability, due to ROI placement, is encoded in the coloring. The left precentral gyrus is an example of a non-anatomically useful ROI that leads to high variability. As shown in (Vavassori et al. 2021), the AF can reach into the precentral gyrus and this ROI forces the AF to go past the precentral gyrus. The ‘rainbow’ coloring observable in the figure shows that the difference in posi-

tioning was large enough to either include or exclude the precentral gyrus in the bundle. As opposed to Fig. 8, this ROI was used as an inclusion criterion, meaning that when being positioned too anteriorly the dissection discarded the streamlines not reaching the ROI. By removing the ROI from the set of rules of the virtual dissection of the AF, not only the bundle would be more anatomically accurate, but a large source of variability would be removed

the source of variability and (iteratively) improve a virtual dissection protocol and that ROIs must be individually evaluated in a bundle-wise manner to truly capture their impact and contribution (this work). These projects demonstrated that virtual dissection protocols cannot be generalized when changing or introducing new variables. The differences between the protocols used for the previously mentioned projects (software, tractography algorithms, teaching methods, etc.) led to heterogeneous results for anatomically similar bundles. These efforts put toward the investigation of WM virtual dissection expose the need for better protocols and anatomical definitions with more rigorous instructions. The following section explores ways the protocol could be optimized to reduce variability or decrease the amount of work from raters. This shows the importance of having an iterative process where decisions are not only justified but tested in “*the field*” to adjust a

protocol to reach the desired anatomical accuracy, reproducibility and workload.

Regions of interest contributions to the virtual dissection protocol

This analysis, which is focused on ROIs, allowed us to determine each ROI intrinsic variability and the impact it has when used in the virtual dissection protocol. The ROIs with the highest positional variability are the temporal stem region. Multiple factors explain why this ROI has such a high variability. First, the starting approximate position for the ROI only refers to the Sylvian fissure anterior to the precentral gyrus ROI (another region with high variability), this leaves the raters to pick a sagittal slice to identify the Sylvian fissure, then since it is an elongated landmark more than one position is appropriate. The rest of the instruction involves a

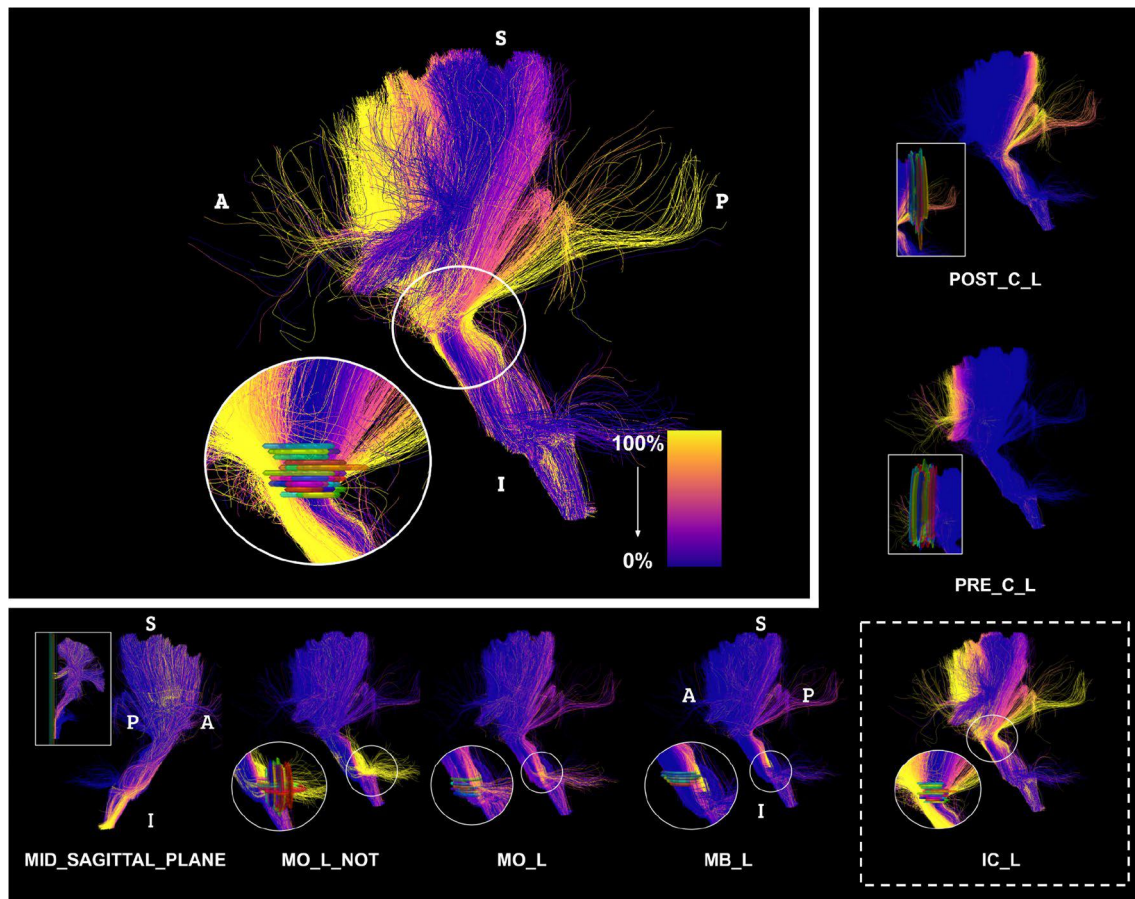


Fig. 10 Mosaic of all ROIs involved in the pyramidal tract virtual dissection. Streamlines variability, due to ROI placement, is encoded in the coloring. The internal capsule is crucial to isolate the pyramidal tract as it is a major bottleneck. However, the difference in positioning and the ROIs' radii leads to large variation (streamlines reaching too anteriorly and/or posteriorly). Positioning disagreement in the inferior/superior axis does not have a major impact, this axis is paral-

lel to the global orientation of the pathway close to the internal capsule. Disagreement in the positioning only has a major impact on variability if that disagreement is perpendicular to the global orientation of the pathways (anterior/posterior or left/right). While not exactly anatomically justified, enlarging the ROI to guarantee that it will most likely encompass the whole internal capsule would decrease the risk of variability

green hypersignal in the temporal lobe (stem of the arcuate fasciculus), this landmark can be observed in multiple slices.

Providing more details could reduce variability, but this leads to the next important point: This variability has little to no effect on the virtual dissection of the arcuate fasciculus. Not only is this ROI difficult to position and has high variability, but it has no contribution to the dissection process. This ROI can be removed from the AF set of rules.

The precentral and postcentral gyri (and the central centrum semiovale, defined using the precentral and postcentral gyri) show high variability too. The precentral gyri have a higher variability for similar reasons to the temporal stem as the initial approximate position involves choosing a sagittal slice to identify the Sylvian fissure and then identify the precentral gyrus (generally along the inferior/superior axis). Despite its high variability, this ROI is useful to the pyramidal tract and corpus callosum virtual

dissection process (as an exclusion rule) and high intrinsic variability only has a small impact on the bundles. However, the same ROI is used (as an inclusion) for the arcuate fasciculus and in this case, the high intrinsic variability does translate to high virtual dissection variability.

In this case, the ROI is anatomically unnecessary (the ROI being anterior to the precentral when in fact the AF should fan into the precentral) and generate high variability. This ROI should be removed from the AF set of rules, but kept for both the CC and PYT. This ROI was initially added due to the fact that spurious streamlines were expected and reaching the most anterior portion of the frontal lobe was a possible solution to avoid this problem. We learned that not only the spurious streamlines we expected were not present, but this additional ROI was increasing variability and became a useful example to the importance of our analysis.

Since there are two recommendations for the AF set of rules (removing the temporal stem and the precentral gyrus ROIs), we verified that applying the new protocol for the AF did not lead to unforeseen results. This could be due to streamlines (that needed both conditions) not being discarded and decreasing reproducibility or anatomical validity. The results were as expected, the removal of the precentral gyrus condition decreased variability and led to an anatomically plausible bundle and the removal of the temporal stem condition did not lead to a significant increase of spurious streamlines. Therefore, both conditions can safely be removed from the protocol.

More detailed analyses and recommendations for each ROI are available in the supplementary materials sections. To summarize, there is a balance between three spectrums: anatomical meaning, effect on the virtual dissection and difficulty to delineate. There is no perfect way to quantify these criteria, anatomical justification must emerge from prior knowledge in neuroanatomy and agreement between experts to justify or not the use of a specific region of interest. However, we believe that the approach used in Figs. 4, 5 helped us inform the “difficulty to delineate” spectrum and Figs. 6, 7 helped us inform the “Effect on dissection” spectrum. The difficulty to delineate applies to each ROI individually, but the anatomical justification and the effect on dissection will vary from bundle to bundle (in which the ROI is used in the set of criteria). Figure 11 summarizes how we evaluated each ROI in the supplementary material to reach useful recommendations to adapt the protocol.

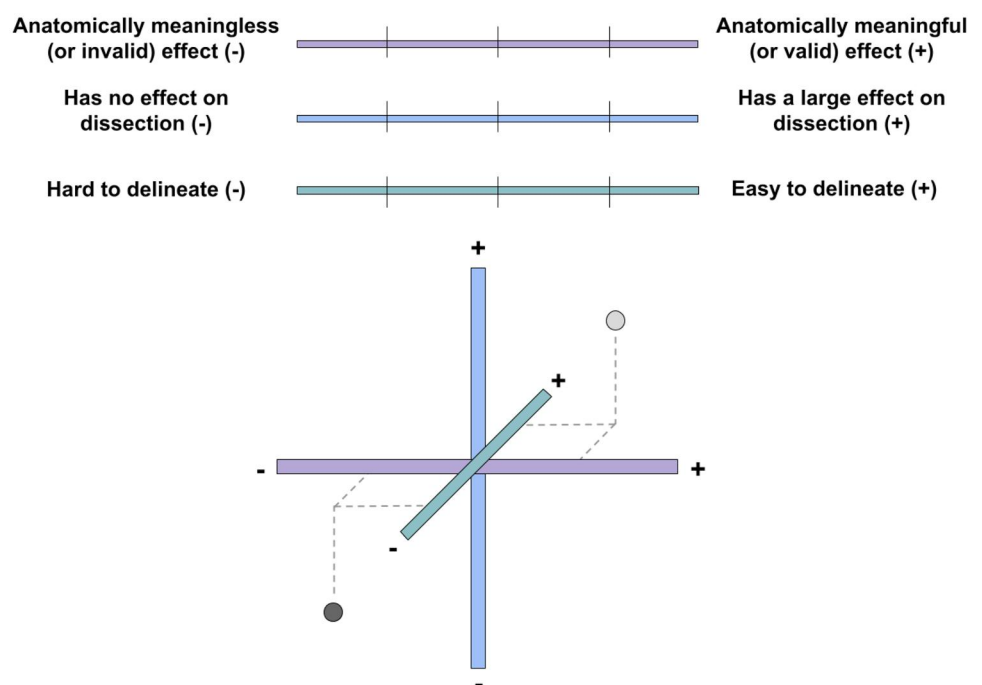
Sadly, this balance cannot be strict and fixed in time. A virtual dissection protocol targeting neurosurgery

applications will not have the same requirements as a protocol design for thousands of healthy subjects, for example. The difficulty to delineate can be different if the raters are experts in neuroanatomy versus nonexperts. However, we believe that the framework presented in this study is required to be able to rank (even if only qualitatively) each ROI on these spectrums and acknowledge the balance to justify their advantage and inconvenience to the protocol. The reader should also know that specific regions of the brain have anatomical variability (i.e., sulcal-gyral variability exists among different individuals or populations). Meaning that even in healthy populations it is possible that some ROI will be inherently harder to delineate for individuals with such anatomical irregularities.

General recommendations

From the multiple projects (including this one) related to practical definitions and virtual dissection reproducibility involving the authors of this project (Maier-Hein et al. 2017; Chenot et al. 2019; Bayrak et al. 2019; Rheault et al. 2020a; 2022a, b; Nath et al. 2020; Schilling et al. 2021a, b) or from other groups (Wakana et al. 2007; Catani and De Schotten 2008; Wassermann et al. 2016; Van Hecke et al. 2016; Warrington et al. 2020; Maffei et al. 2021) led us to propose some recommendations for the definition of WM pathways’ practical extraction. These recommendations are also inspired by guidelines of the EADC-ADNI Harmonized Protocol for manual hippocampal segmentation (Frisoni et al. 2015). These general guidelines are provided to help future protocols design, some adaptations could be required

Fig. 11 In the current study, three main spectrums describe each ROI in the virtual dissection protocol. While mostly a qualitative ranking, it is important to consider all three when providing recommendations. The extreme cases are easy, such as a ROI easy to delineate, with large and meaningful/valid effect on the dissection (and the opposite). ROI with no effect or invalid effect should be removed, ROIs that are hard to delineate with large effect should have their instructions reviewed to reduce variability



for young children, older adults, lesioned brains or for data in high-resolution, from ultra-high gradients, from 7 Tesla MR scanner, etc. However, it is likely that virtual dissection of such datasets would need a targeted protocol with instructions tailored to the population and that these general recommendations would still be somewhat relevant for its design. More details for each recommendation are available in the supplementary materials.

Dissection in standardized space

Virtual dissection protocol should always be performed in a standardized space (e.g., MNI or AC-PC aligned). The reason for this recommendation is simple: landmarks do not look the same depending on the brain orientation due to slicing. Relative spatial location is easier to interpret (by humans or by computers in MNI/AC-PC align space (medial to, lateral to)).

Clear instruction on the software does help raters

While the description of regions of interest is crucial, a detailed presentation of the software is likely as important. Presenting functionalities, tips to prevent major pitfalls, avoiding crashes and saving time all are important to reduce the mental workload of a rater. Written text is a good medium for a reference guide, but step-by-step instruction (or better, a video) is a good way to increase reproducibility.

Virtual dissection workflow

Another way to reduce variability is to establish a workflow for the tasks. Organizing the input, the order in which the data should be loaded in the software, and requiring a naming convention for all ROIs and BOIs are all solutions to reduce the workload of raters.

Getting in the zone

While the accuracy of a virtual dissection does not significantly change over time, each rater workflow (software manipulation, files structure, details about the protocol, file-name convention) needs some time to get up to speed. It is good practice to set time to focus and perform a few virtual dissections at the time and to always complete a dataset before stopping.

Being too strict is worse than being permissive

Using very small ROIs or using too many exclusion regions is not only harder for raters, but it leads to poor results. Using permissive and easier to locate ROIs leads to good results. Combining multiple ROIs will facilitate the process

of virtual dissection. A good balance of inclusion and exclusion is a better approach than relying mostly on one or the other, but using too many ROIs can increase the workload of raters.

Using other ROIs as landmarks

This is generally a good practice to facilitate and accelerate the virtual dissection process. By providing a lot of details on specific ROIs, the subsequent regions should become easier to locate as they reference previously drawn ones. However, this is true only if used for ROIs that are highly consistent. A region with high variability will ‘transfer’ its variability to any other region referencing it.

Generating more than one ROI from the same landmarks

Only re-use ROIs if they have the same impact, if not a new ROI may be needed. For example, if a ROI is used twice, once as inclusion and once as exclusion, and they lead to different effects on their respective dissection, it could be more reproducible and easier for raters to create two separate ROIs from the same set of instructions.

Providing ranges

Each ROIs should have an approximate initial location and an approximate size to accommodate the anatomical variability of subjects. When in a standardized space, a range of slices can be provided to facilitate location. The exact value should be avoided, raters should be instructed to start from an approximate location and scroll until something specific is located.

Locating landmarks

It appears that instructions using the diffusion-encoded color (DEC) map (or RGB map) are easier to follow than the ones using only T1. This is likely because DEC contains hyper-signal of three main colors, much easier to describe and locate than a general area of ‘gray’ on the T1. It is worthy of mention that composite T1-DEC map images could provide similar advantages and that DEC obtained from DTI or CSD could have different contrast quality, which could influence the raters’ decisions. It also appears that descriptions that involve multiple orientations at once are difficult (axial, coronal sagittal). Landmarks that are perpendicular to the slicing orientation should be prioritized to limit switching from one plane to the other (in the software).

Optimal ROIs design

When using the inclusion rule for a stem/core, the positioning along the pathway is less important, the focus should be targeted at the perpendicular positioning (perpendicular to a strongly oriented core). A larger diameter is likely better than a smaller one. When using the exclusion rule, bigger is often better as long as you do not accidentally cut in half the pathway somewhere else (e.g., middle sagittal plane to prevent commissural streamlines accidentally blocking the brainstem). For exclusion rules, adding a few millimeters in the opposite direction of what is supposed to be stopped provides a good margin of error to reduce variability.

Increasing complexity

The virtual dissection protocol should start with basic descriptions of the bundles of interests, a summary of the steps that will follow and a general overview of all regions that will be drawn. ROIs' instruction should start with a fast (and less accurate) set of instructions to delineate the region followed by a slower (more accurate) detailed instruction.

Cleaning ROIs

If cleaning ROIs are needed (not anatomically meaningful, but simply to clean outliers from tractography limitations), they should be described and justified, follow the previously listed recommendations and all cleaning ROIs should be accounted for. It should not be left to the raters to decide how big, how many, where cleaning ROIs are needed. If the protocol covers a range of acquisition quality, local modeling or tractography algorithms, the cleaning ROIs could be tailored for each type to avoid generic description and inadvertently increase variability.

Transfer of knowledge: why are these projects and recommendations useful? (opinions from co-authors)

For a neuroanatomist?

More than any other protagonist of the current study, the neuroanatomist faces the crucial question of the definition and denomination of a given white matter pathway, especially when confronted with the question of its virtual dissection in tractography (Mandonnet et al. 2018; Panesar and Fernandez-Miranda 2019; Porto de Oliveira et al. 2021; Vavassori et al. 2021). There is a serious lack of consensus on how to describe the fascicular organization of the white matter of the human brain, mainly because we do not yet understand it well enough. A commonly accepted definition of a given fascicle will only be possible by multiplying the

definition criteria. The Tractostorm projects and the resulting recommendations show, both qualitatively and quantitatively, how the number and position of ROIs that will characterize part or all of a bundle's path through white matter must be part of this consensus definition in the future. The process needed to reach such a consensus, likely through iteration, will likely need tractography and virtual dissection.

For a neurosurgeon?

The results of the current work highlight the difficulty in achieving optimal intra-rater and inter-rater reproducibility, which may in part contribute to the discordant findings in neurosurgical studies that report on the utility of tractography in the perioperative management of brain pathologies. Understanding these drawbacks is primordial to correctly interpret the results of studies involving bundle dissection and characterization. Nevertheless, the effort made by these projects in establishing reproducible dissections opens the door to a framework that has the potential to shift the currently imperfect paradigm towards a unifying methodical approach and prompt identification of WM bundles.

Despite the studied protocols targeting healthy subjects, the current series of studies have direct implications in the selection, presurgical evaluation, operative and postoperative management of patients affected by various neurological diseases. Being able to accurately characterize WM tracts has the potential to directly influence patient care. A better understanding of WM pathways segmentation (even in healthy subjects) is a step forward. Ultimately, this work led to a better understanding of how tractography can be interpreted (with all of its challenges and biases in virtual dissection).

When dealing with distorted neuroanatomy it is crucial to remember that a protocol has to be modified and therefore its assessed variability will not be the same. For example, removing any region of interest in the area affected by a pathology (e.g., tumors). The same goes for modifying the processing algorithm from the initial protocol to account for a pathology. For example, if instead of a whole brain tractography using a WM mask, a tractography was seeded from BOLD functional MRI activation regions or mapping from direct cortical electrical stimulation and tracked in a fractional anisotropy (FA) map a specific virtual dissection protocol produced entirely different results. While understanding normative neuroanatomy is fundamental to safe neurosurgical practice, any translational knowledge to neurosurgery must be carefully evaluated (Yang et al. 2021).

For a data scientist?

The results of this work are immediately useful for data scientists. A data scientist may be one who 'creates' the tools

(i.e., the software, algorithms, and dissection protocols) or one who ‘utilizes’ these tools and processes. For those who create the tools, the process of targeting a pathway must be supplemented with clear instructions, with emphasis on the steps that are more critical for reduced anatomical variability. It is necessary to understand sources of variability to reduce it, which will ultimately lead to the identification of smaller effect sizes when studying changes across pathways, subjects, or time. Not only can these results be used to continually improve manual protocols, but also used for creating automated or semi-automatic segmentation protocols. Methods based on regions in a standard space, or those developed to learn streamline or region placement, must take into account the lessons learned here to enable robust and reproducible segmentation and subsequent analysis. Training and validation of semi- or fully automated algorithms must consider the variability inherent in generating the ‘ground truth’ bundles upon which it is trained and compared against. Finally, those that utilize the tools must be aware of variance due to intra/inter-sites, intra/inter-raters, and variability inherent within the methodology to draw informed conclusions about the studied pathways.

Conclusion

This work shows what is driving differences between virtual dissections. Our analysis shows that despite high variability, manual delineation can have, at the same time, low impact on some bundles and high impact on others. These results demonstrate the importance of not only quantifying variability but that it must be done for every combination of regions of interest and pathways.

In this study, we propose a framework to evaluate the individual contribution of regions of interest in a virtual dissection protocol. This information is complementary to prior work from the same group of authors and contributes to better design anatomical definitions (practical or theoretical) of white matter pathways. Our in-depth investigation of the source of variability provides insight into more efficient ways to transmit knowledge related to structural neuroanatomy and tractography.

Our current and prior contributions were incremental and tedious and there are many technical challenges still to be resolved, but we paved the way forward to a more coherent approach to convey knowledge in neuroanatomy and tractography. Having more efficient communication between groups with different expertise is crucial in a multidisciplinary field and we hope that our work brings us closer to a consensus on how to define major white matter pathways and reproducible/replicable science.

We hope to continue our close collaboration with neuroanatomists and neurosurgeons to further improve both

theoretical and practical definitions of white matter pathways. A natural extension of this work is to design, develop and validate other bundles. We believe our current, and previous, work will facilitate our, as well as others, effort toward this natural extension.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00429-022-02518-6>.

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Data availability This study has been reviewed by the Internal Review Board (IRB) of Vanderbilt University (#211156). All raters agreed to collaborate for the data gathering phase. The datasets generated before the study and the obtained files from the raters (analyzed during the current study) are available on Zenodo: <https://zenodo.org/record/6473141>.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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Supplementary materials

Pyramidal tract - Internal Capsule (IC)

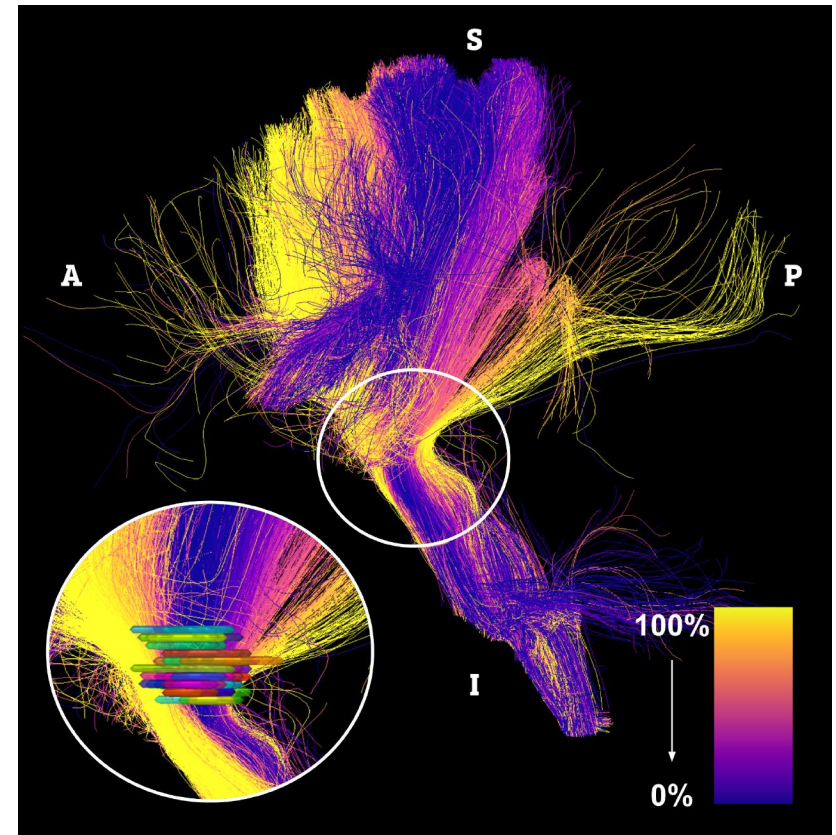
The internal capsule (IC) region of interest specifically targets the genu, which is the point of passage for the motor and somatosensory that originate from the cortex. This specific region acts as a stem in which the pathway of interest must go through.

Raters were expected to identify the ROI using an approximate location (in the I/S axis) from which they could locate the thalamus. Using the barycenter of the thalamus, they had to find a hypersignal in the b0 image (due to high myelination on top of a blue region on the RGB).

The identification of the thalamus and its approximate barycenter will lead to variability in the I/S axis. When the IC region itself is found, the genu of the IC can be difficult to identify due to the poor contrast of the b0 and the diffuse blue region on the RGB. This leads to variability in L/R and A/P directions as well as variation in the size (radius) of the ROI to draw to encompass the IC.

Deviations from the true IC genu lead to a decrease in density of streamlines and often a loss of fanning in the pre/post-central (medial regions). Large deviation results (positioning the ROI in the anterior or posterior limb of the IC) can result in segmenting a different pathway.

The difficulty to identify the I/S axis did not lead to major variability, but the L/R and A/P axis. Increasing the recommended size of the ROI could reduce variability. Instead of trying to exactly identify the genu of the IC, identify the IC as a whole and relying on other regions (already part of the protocol) to constrain the pathways would improve reproducibility.



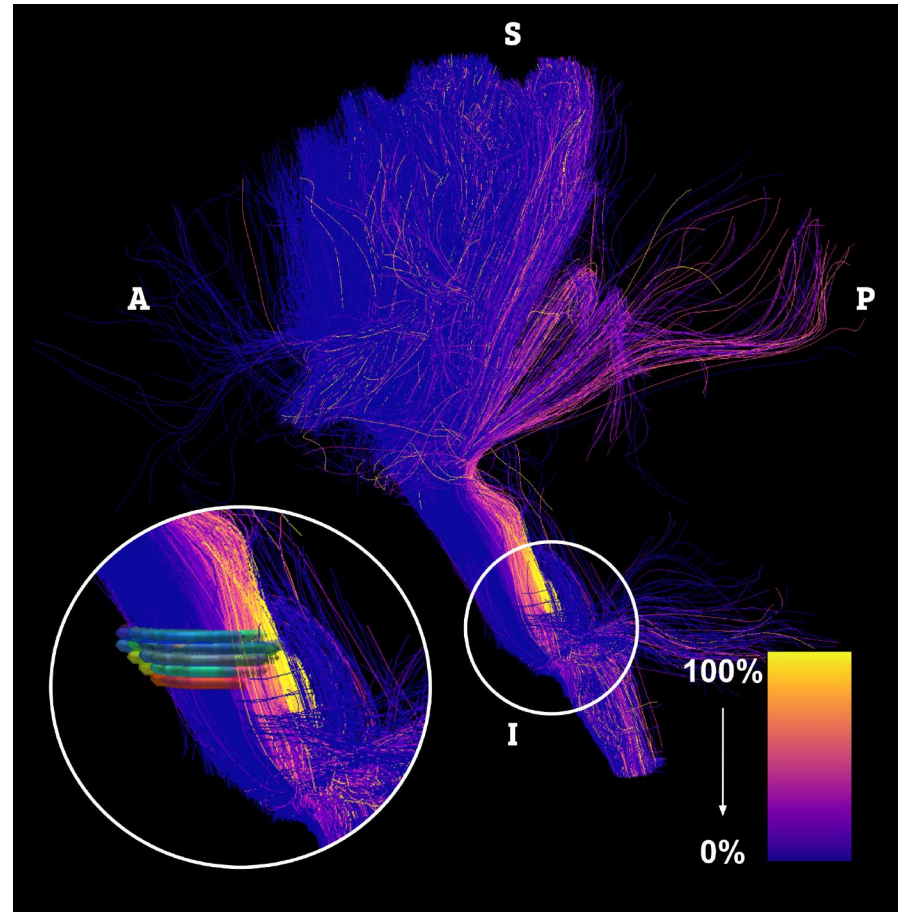
Pyramidal tract - Midbrain (MB)

This ROI is used to discriminate the PYT (motor pathway) located in the anterior part of the midbrain from the lemniscus tract (sensory pathway) located in the posterior part of the midbrain.

An approximate position (in the I/S axis) was recommended as a starting point in the brainstem, then the exact position of this ROI was located using the RGB color map and hypersignal in blue.

Since this ROI is located in between the IC and MO, there was relatively small variability in the I/S axis and the most anterior blue hypersignal was clearly distinguishable. The ROI had to encompass the blue hypersignal, this variable was clear to determine. Raters located and drew the region with high consistency.

Variation in position and size had very small effects on the resulting bundles. Overall, the spatial extent of the bundle is mostly unchanged by the small disagreements between raters' ROI.



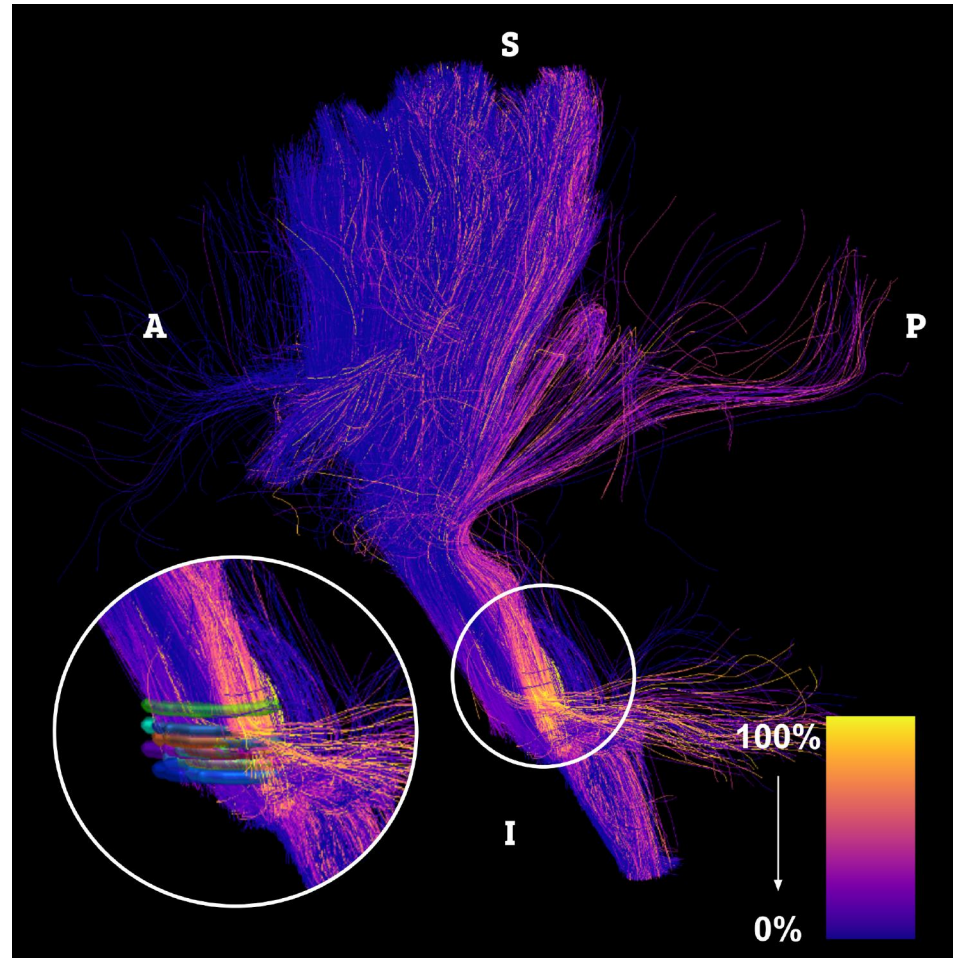
Pyramidal tract - Medulla Oblongata (MO)

This ROI is used to select fibers entering in (or going into) the medulla while not passing through the cerebellum. This ensures the selected pathways do not end in the pons and continue down the brainstem.

An approximate position (in the I/S axis) was recommended as a starting point in the brainstem. To position the ROI in the axial plane the RGB was used to locate where the white matter of the medulla oblongata (rather in blue) separate from the white matter of the cerebellum. Then, the ROI is centered on the blue signal located in the anterior part of the Medulla Oblongata.

Due to the clear landmark and approximate starting position, the positioning of the ROI was consistent. Only small effects can be observed on the resulting bundles. Overall, the spatial extent of the bundle is mostly unchanged by the small disagreements between raters' ROI.

However, there is a potential algorithmic pitfall for this region (which did not occur in our study). Tractography often abruptly ends just after the medulla (due to partial volume effect and a bend in the spinal cord). This means that while tens of thousands of streamlines can reach a ROI positioned just above the medulla, only a few thousands will likely reach a ROI positioned just below it. Anatomically speaking, the pathway is expected to go beyond the medulla, but there is a balance between tractography reconstruction and anatomical accuracy. Positioning this ROI too low will result in a major loss of density, which will affect the spatial extent and likely the lateral fanning of the PYT.



Pyramidal tract - Mid-sagittal plane

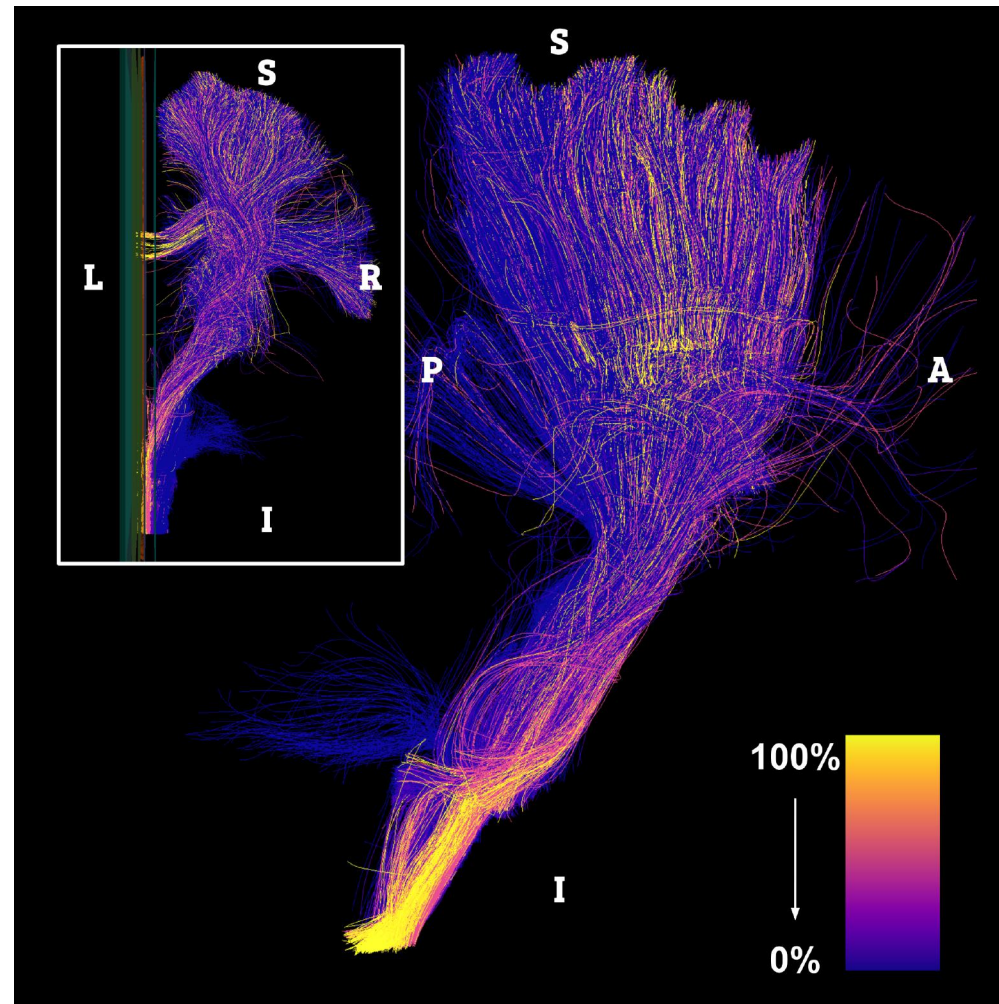
This region aims to prevent spurious streamlines from switching paths into the corpus callosum (outliers) and into the other hemisphere.

Raters were expected to identify the interhemispheric fissure and select the entire plane as a ROI.

There was little potential confusion for this region, the observed variability was only in the L/R axis. However, by selecting the whole plane if raters selected a slice too lateral it is possible to cut the brainstem. In the current dataset, this kind of error was rarely observed and only had a limited impact.

To avoid this potential source of variability, it would be wise to use a hand-drawn ROI instead of using an entire plane. In the case of non-AC-PC aligned datasets, this small modification could avoid a major loss of streamlines by accident.

As shown in the figure, the variability comes mainly from the brainstem and only affects a few commissural outliers. While useful to remove undesired outliers, the ROI creates variability in an unexpected region.



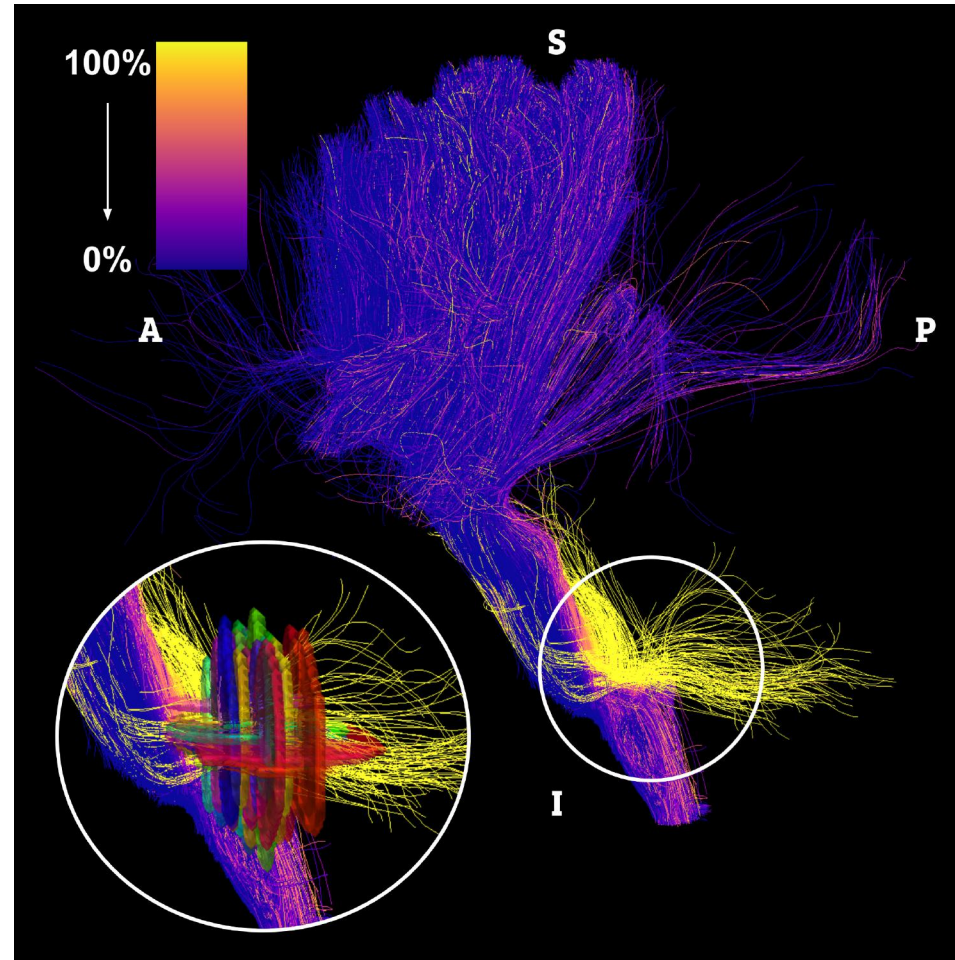
Pyramidal tract - Cerebellum cleaning

This region aims to prevent spurious streamlines from entering the cerebellum and avoiding contamination from the lemniscus tract (sensory pathway) located in the posterior part of the midbrain.

This is the only '2D' ROI of the protocol (two intersecting disks). The raters were asked to draw a large (axial) disk behind the MO-ROI on top of a blue signal on the RGB (to block the posterior part of the midbrain). Then, a second (coronal) disk had to be drawn to cut off the entrance of the cerebellar peduncle without affecting the streamlines reaching down to the spinal cord.

While the intent of the ROI is intuitive, such an exclusion region in the brainstem can be risky. If too large or placed too low, the coronal disk is at risk of intersecting the lower brainstem. This would severely impact the density and spatial extent of the bundles. Similarly, if the axial disk is too large or placed too low, it comes near the anterior portion of the pathway and is a risk of preventing the streamlines from reaching the lower brainstem.

For the description, interpretation and placement of the ROI, using two independent disks instead of presenting it as a single ROI made of two intersecting disks would likely decrease variability. It would at least reduce confusion, raters feedback showed that this ROI was perceived as difficult. However, as seen in the figure, the ROI is a low source of variability and the impact of raters' variation only impacts the desired area (cerebellum). Despite its potentially high risk, the ROI was well executed by most raters.



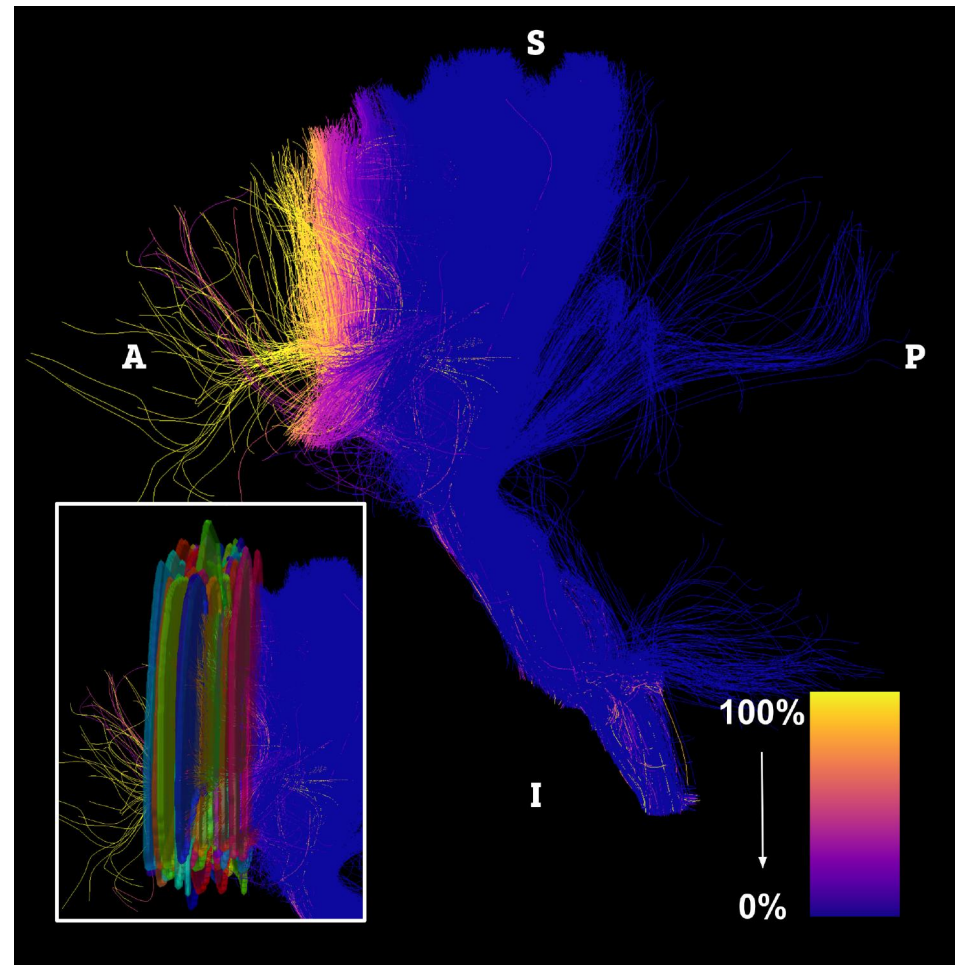
Pyramidal tract - Precentral gyrus

This region aims to prevent streamlines from reaching into the frontal lobe; it acts as an anterior limit to the bundle when reaching the cortex. The ROI is a large planar drawing blocking any streamlines going too much forward.

Since the precentral gyrus has a complex orientation, it would have been difficult for raters to accurately identify the whole gyrus. Instead, the raters had to identify the most anterior limit of the precentral gyrus by first locating where Sylvian fissure starts and then scrolling medially until the lateral sulcus divides both the frontal lobe and temporal lobe. Finally, raters had to draw the planar ROI where the precentral gyrus generally runs in the I/S axis.

The precentral ROI shows little variability in the L/R and I/S axis. However, major variation in the A/P axis can be seen in the figure, despite raters locating the ROIs across more than 20mm distance only small impacts can be observed. Variations in size and shape have a low impact on the final bundles.

As an exclusion ROI, adding an extra margin of error (moving it more anteriorly) would increase reproducibility. However, this ROI is used by the corpus callosum (as an exclusion) and arcuate fasciculus (as an inclusion) and thus this solution may not be optimal for the other bundles.



Pyramidal tract - Postcentral gyrus

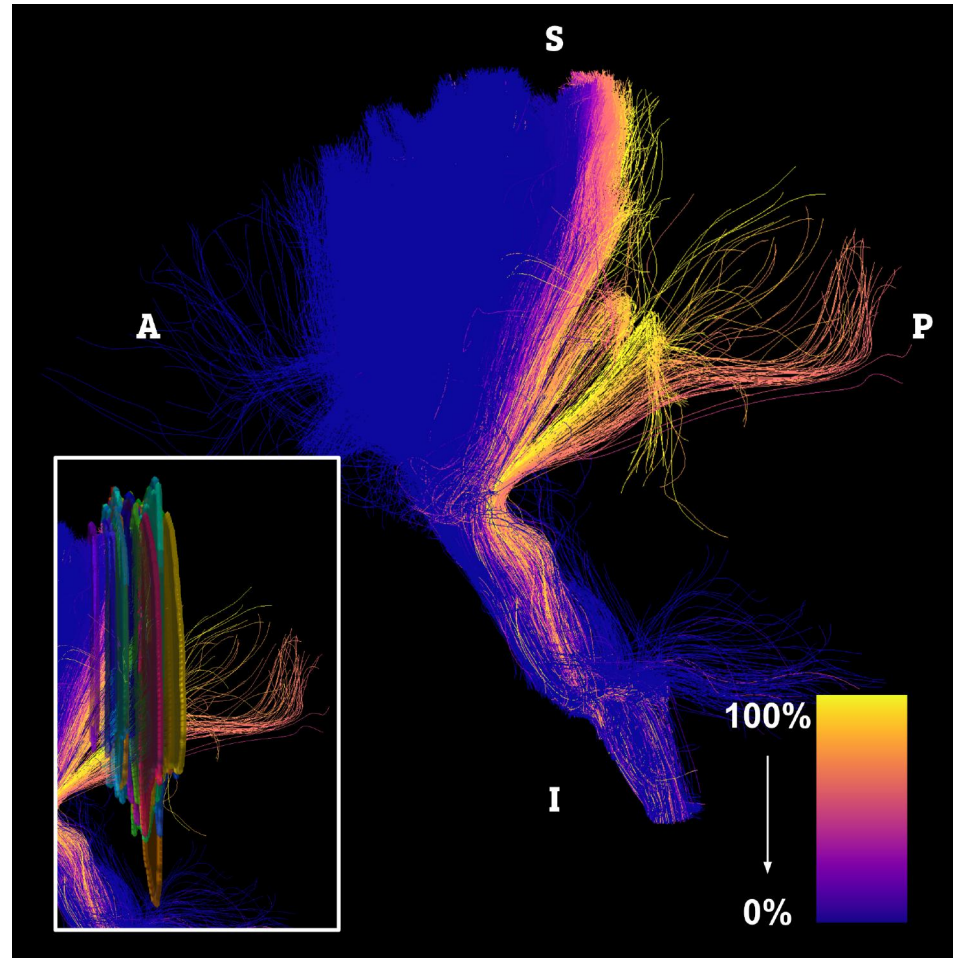
This region aims to prevent streamlines from reaching into the parietal/occipital lobe; it acts as a posterior limit to the bundle when reaching the cortex. The ROI is a large planar drawing blocking any streamlines going too much backward.

Similar to the precentral gyrus, requesting an accurate segmentation of the whole structure would be difficult. To avoid this a single planar ROI had to be positioned at the most posterior portion of the postcentral. Raters were asked to identify the Rolando sulcus, known as the deepest (most medial termination) sulcus seen on the axial slice and that includes the omega shape. Then, the postcentral gyrus was easy to identify (one gyrus back) and from this position, they had to draw the ROI.

The postcentral ROI shows little variability in the L/R and I/S axis. However, major variation in the A/P axis can be seen in the figure (similar to the precentral, around an average 20mm distance). Only small impacts can be observed, variations leading to major differences are rare.

Just like adding an extra margin of error (moving it more posteriorly) could increase reproducibility for the precentral ROI, this could work for the postcentral ROI.

However, this ROI is used by the corpus callosum (as an exclusion) and arcuate fasciculus (as an inclusion) too. In the first version of Tractostorm, it was observed that some raters drew this region too large (reaching as far down as the brainstem) and led to streamlines being blocked from the lower brainstem. Likely due to the description and demonstration from the online course, this error was avoided by the rater. This shows how the information is shown in a didactic course (even small details) leads to an increase in reproducibility.

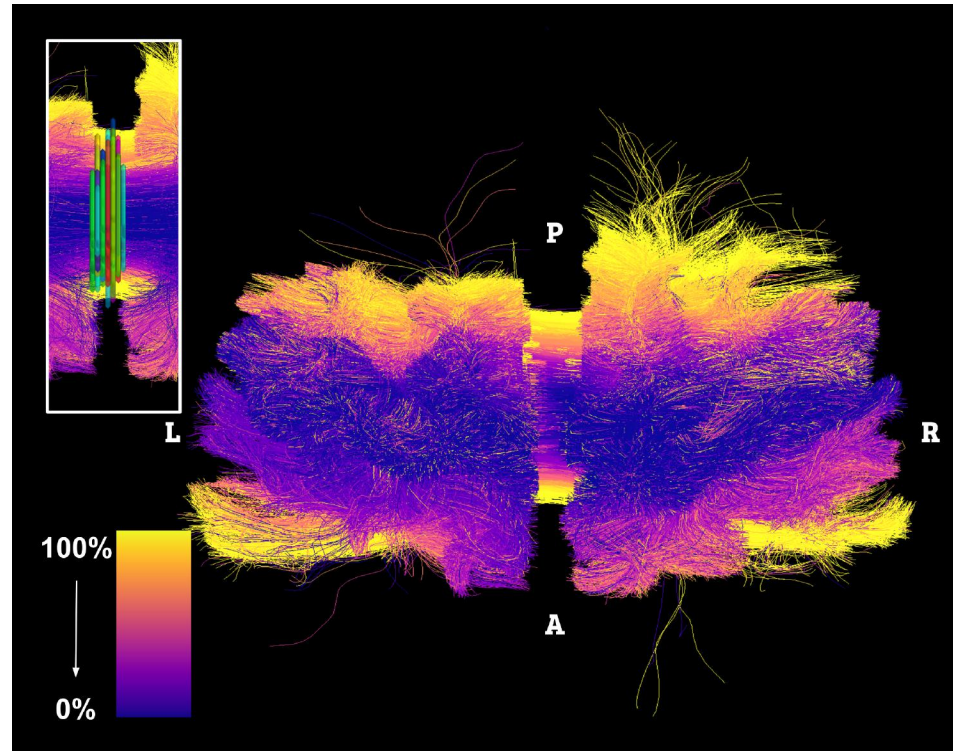


Corpus callosum - Central corpus callosum

This ROI aims to approximate the mid-anterior, central and mid-posterior regions of the corpus callosum. This was made to target streamlines that fan into the precentral and postcentral gyri.

Raters were requested to identify the small rectangular region on the mid-sagittal slices, in between the precentral and postcentral ROI drawn previously. The region was expected to be 20-30mm long rectangle with minimal curvature in the corpus callosum, excluding the genu and the splenium)

The intended 'core' of the bundle was adequately segmented by most raters. Variations in the L/R axis had very little effect. The A/P axis position of this ROI is first approximated by the 'center' between the precentral and postcentral gyri regions, this leads to a bias. Raters that estimated these ROIs too forward or too backward will not only constrain the bundle too little (either anterior or posterior), but will also likely estimate this ROI in the wrong location. Variations in the A/P axis resulted in a bundle that overflowed anteriorly or posteriorly. It was also observed that missed voxels in the I/S axis due to the region being too small resulted in hundreds of streamlines being missed.



For the I/S axis variation, the easiest solution is to enlarge the region to make sure the whole thickness of the corpus callosum is captured. As for the A/P axis variation, the solution is much more complex. First, the ROI could also be enlarged in this axis, but to constrain the bundle a good segmentation of the precentral and postcentral is needed. This is a careful balance, by enlarging this region of interest, more of the 'responsibility' is on the precentral and postcentral regions. As seen in the figure, variations have a large visual impact. However, the obtained bundle generally conserved a stable density, but the spatial extent varied drastically.

Corpus callosum - Lower axial limit

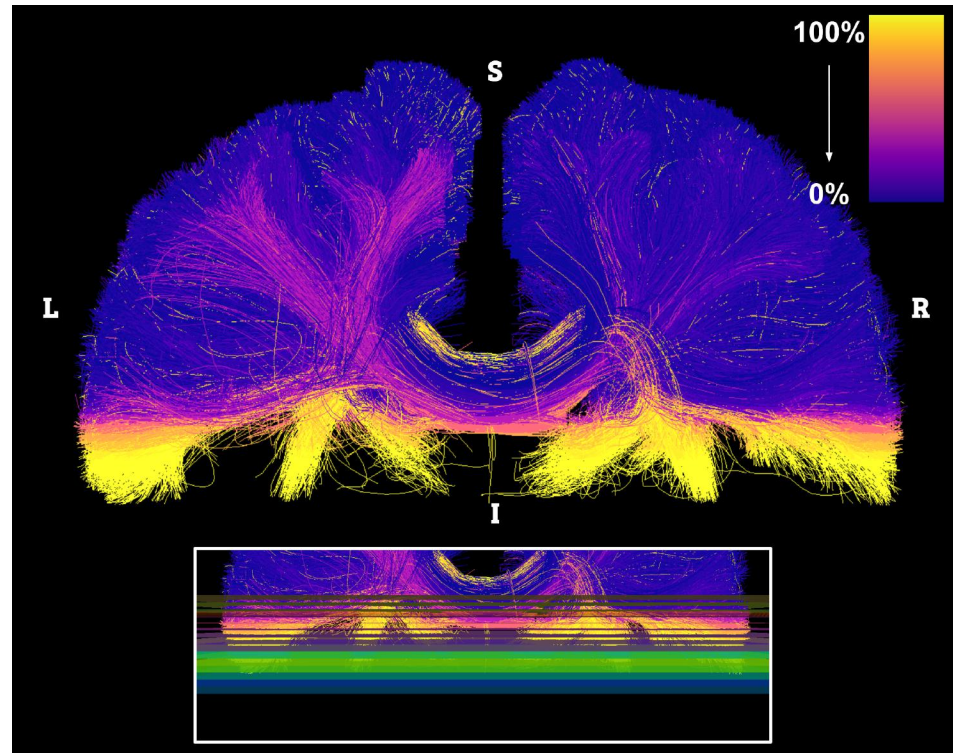
This region aims to prevent spurious streamlines from switching paths into the pyramidal tract (outliers) and from reaching into the deep nuclei.

Raters were expected to identify the lower axial limit using another ROI as a landmark (body of the corpus callosum). The plane is supposed to be located below the ventricle or the Sylvian fissure, whichever is lower.

Depending on the coronal slice selected to evaluate the position of that limit, a lot of variability occurs in the I/S axis. When placed “too high” the ROI would intersect fanning in the lateral portion of the corpus callosum.

To avoid this potential source of variability, it would be wise to use a hand-drawn ROI instead of using an entire plane. By limiting the ROI to only the region immediately below the ventricles, this would preserve lateral fanning. In the case of non-AC-PC aligned datasets, this small modification could avoid a major loss of streamlines by accident.

As shown in the figure, despite high variability in the I/S axis only a small portion of the corpus callosum is impacted. The ROI could also be drawn lower by adding a few millimeters as a margin of error. This would still eliminate the outliers switching into the pyramidal tract while reducing the variability observed at the cortex.



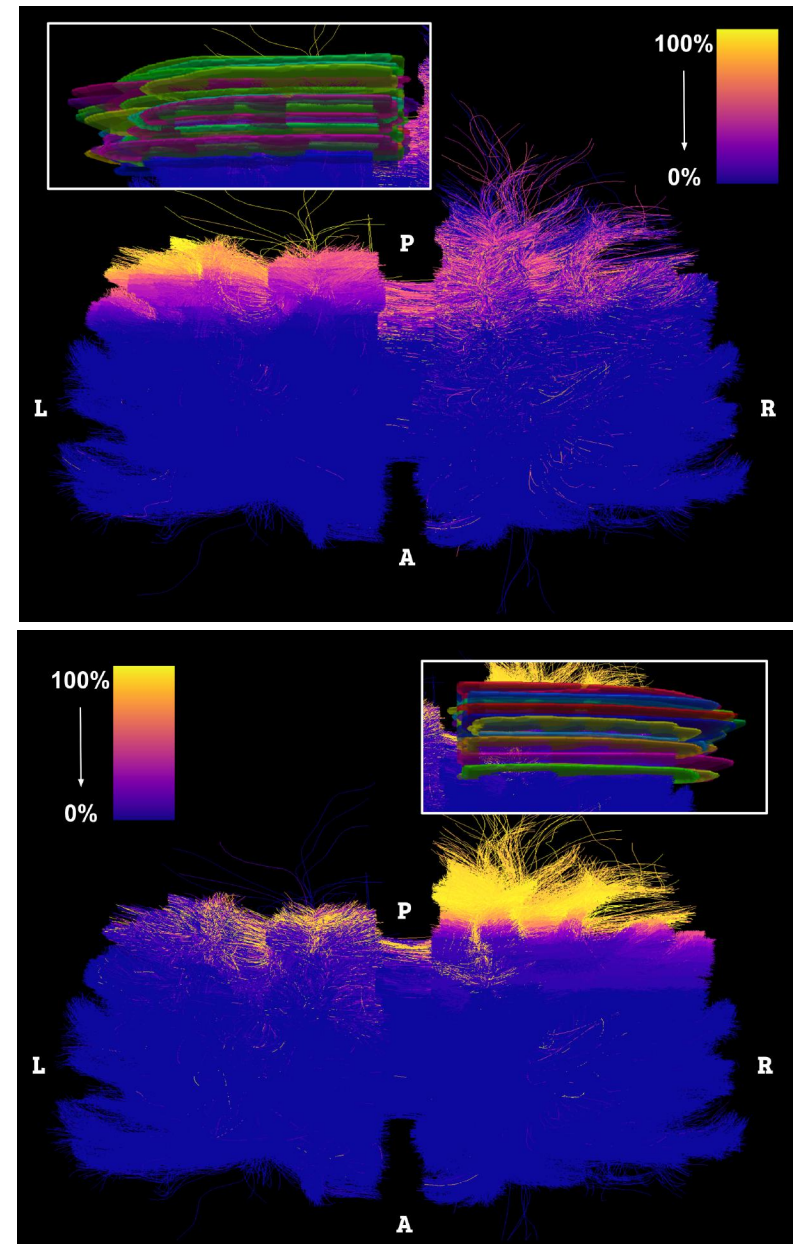
Corpus callosum - Precentral gyrus (L/R)

These regions aim to prevent streamlines from reaching into the frontal lobe; it acts as an anterior limit to the bundle when reaching the cortex. The ROI is a large planar drawing blocking any streamlines going too much forward.

The precentral gyrus ROI on the left is the same as the one used for the pyramidal tract. The goal was to reuse as much as possible to decrease the total duration of the protocol. The ROI on the right is obtained by following the same instructions, but applied to the right hemisphere (for more details, see above).

Similar to the pyramidal tract, variability in the I/S axis causes little variation in the final bundles. Despite raters locating the ROIs across more than 20mm distance, only small impacts can be observed.

The corpus callosum dissection uses the same ROIs (and its counter-hemispheric analog) as an exclusion ROI (just like the pyramidal tract). Adding an extra margin of error (moving it more anteriorly) would increase reproducibility without decreasing the anatomical accuracy of the segmented bundles.



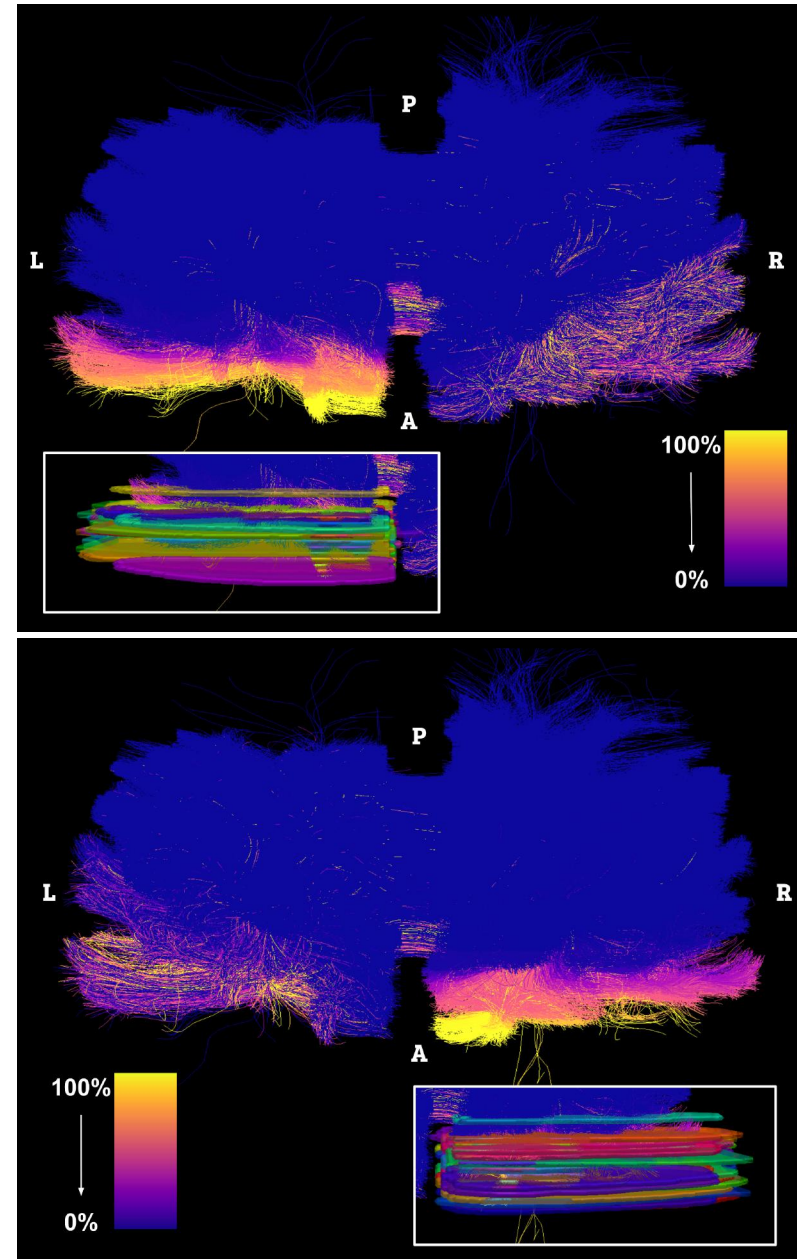
Corpus callosum - Postcentral gyrus

This region aims to prevent streamlines from reaching into the parietal/occipital lobe; it acts as a posterior limit to the bundle when reaching the cortex. The ROI is a large planar drawing blocking any streamlines going too much backward.

The postcentral gyrus ROI on the left is the same as the one used for the pyramidal tract. The goal was to reuse as much as possible to decrease the total duration of the protocol. The ROI on the right is obtained by following the same instructions, but applied to the right hemisphere (for more details, see above).

Similar to the pyramidal tract, variability in the I/S axis causes little variation in the final bundles. Despite raters locating the ROIs across more than 20mm distance, only small impacts can be observed.

The corpus callosum dissection uses the same ROIs (and its counter-hemispheric analog) as an exclusion ROI (just like the pyramidal tract). Adding an extra margin of error (moving it more posteriorly) would increase reproducibility without decreasing the anatomical accuracy of the segmented bundles.



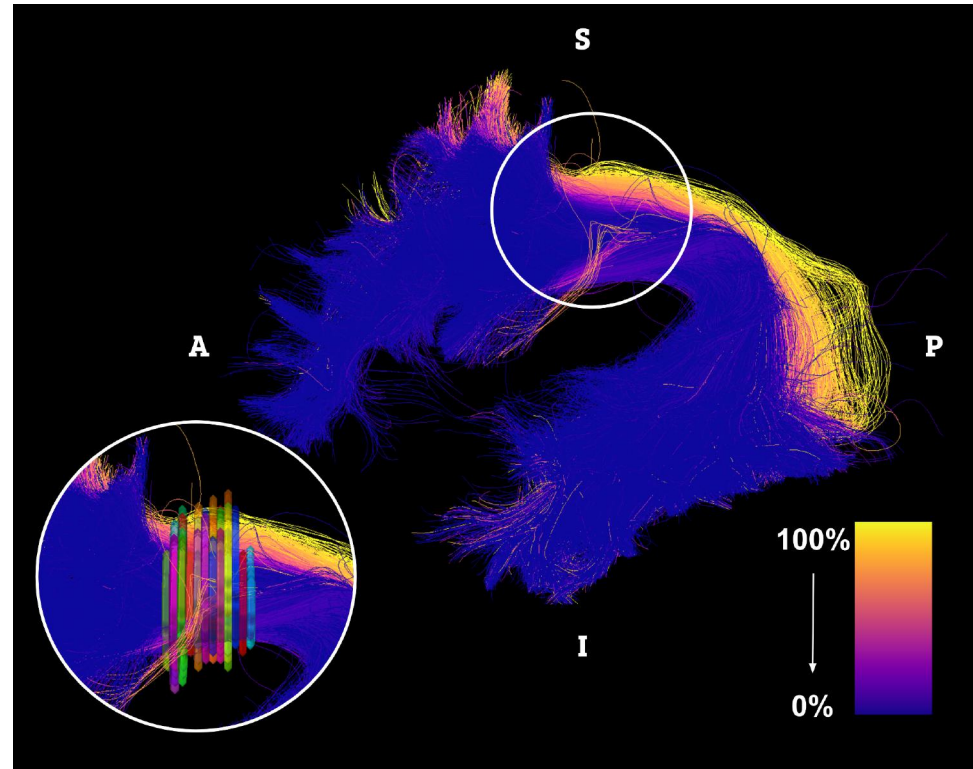
Arcuate fasciculus - Centrum semiovale

This region aims to force streamlines to funnel into the portion of the centrum semiovale close to the precentral and postcentral gyri. It acts as a bottleneck for the bundle before fanning into the frontal lobe. The ROI is a large disk that targets the core/stem of dorsal association pathways.

The approximate location of the ROI was determined by using the center point (in the A/P axis) between the precentral and postcentral ROI delineated earlier. Then, a large triangular green hypersignal on the RGB had to be located (generally below the curve of the corpus callosum pathway). A large disk encompassing the arcuate fasciculus core/stem (15-20mm) had to be positioned at the center of the green hypersignal.

Variations in the I/S axis caused little impact on the final bundles. However, the variations in the L/R and I/S locations as well as the disk radius differences are causing variability at the periphery of the curve into the temporal lobe while the core/stem is correctly selected. This has a low impact on the overall volume and no impact on the distribution of streamline density in the bundle.

For this particular region of interest, increasing the size is likely not a desirable alternative since it aims to target the stem of the pathways and disentangle the arcuate from other association pathways (e.g. superior longitudinal fasciculus). Focusing on better identification of the precentral and postcentral gyri (and consequently the center point in between) would provide a better starting position of the raters to identify the green hypersignal on the RGB. More details are needed to identify the center of the green hypersignal. More specifically, instructions on how to 'follow' the curvature of the corpus callosum pathways to give a superior limit. Since the medial limit of the arcuate fasciculus plane (describe below) is easier to describe, it should be used as a reference (instead of using the centrum semiovale as a reference for medial limit)



Arcuate fasciculus - Medial sagittal limit

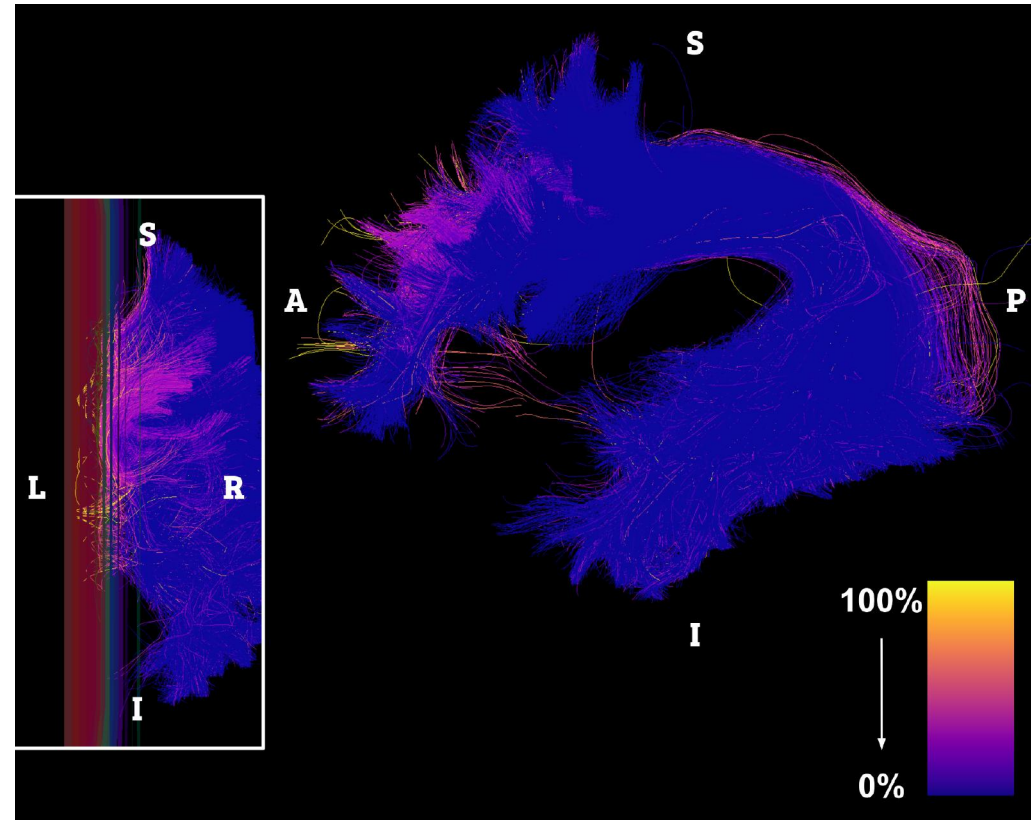
The medial sagittal limit acts as a constraint to avoid streamlines from the superior longitudinal merging into the arcuate fasciculus. This region also prevents spurious streamlines from switching paths into another bundle such as the corpus callosum (outliers)

Raters were expected to identify the ROI by using a previously drawn region (centrum semiovale) as a slice reference and use the outer edge of a green region on the RGB as a limit.

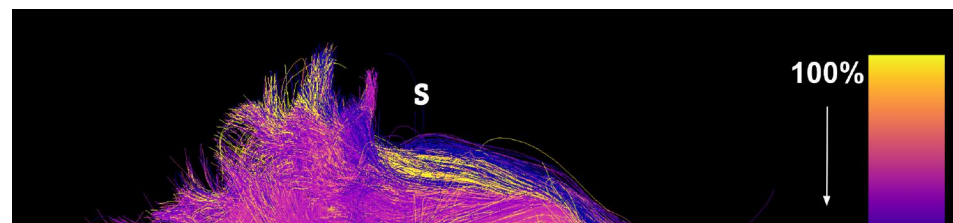
Since the plane selection strongly relies on another ROI, the previous decisions will have repercussions. For example, the choice of a slice to visualize the green signal will impact the outer edge limit. The green signal can mix with another color (e.g. blue with the PYT) and so determining the most medial position of the green signal is not trivial.

The impact of variations in the L/R axis (from the centrum semiovale segmentation) was limited. However, determining with accuracy the end of the green region medially was the main source of variation. Personal interpretation of what is and is not “green” probably was the source of confusion. Despite the variation in the L/R axis, the impact on the final bundles was limited.

Choosing a slice too laterally would lead to a major difference in the bundle as it would intersect the centrum semiovale and effectively cut in half the arcuate fasciculus. Adding an extra few millimeters medially as a safe margin of error would reduce variability while keeping the same purpose.



Arcuate fasciculus - Temporal lobe entrance



This ROI is designed to approximate the entrance to the temporal lobe and ensure that reach into the temporal lobe (and thus achieve the U-shape of the arcuate fasciculus)

The raters were asked to locate a slice where the temporal lobe is clearly distinguishable approximately on the same coronal slice as the postcentral gyrus ROI. Raters were also asked to use the lower axial limit ROI as a landmark to ensure the temporal lobe entrance is strictly below the Sylvian fissure. The ROI was described as a large square/circle ROI of approximately 25-35mm in radius to encompass the temporal lobe.

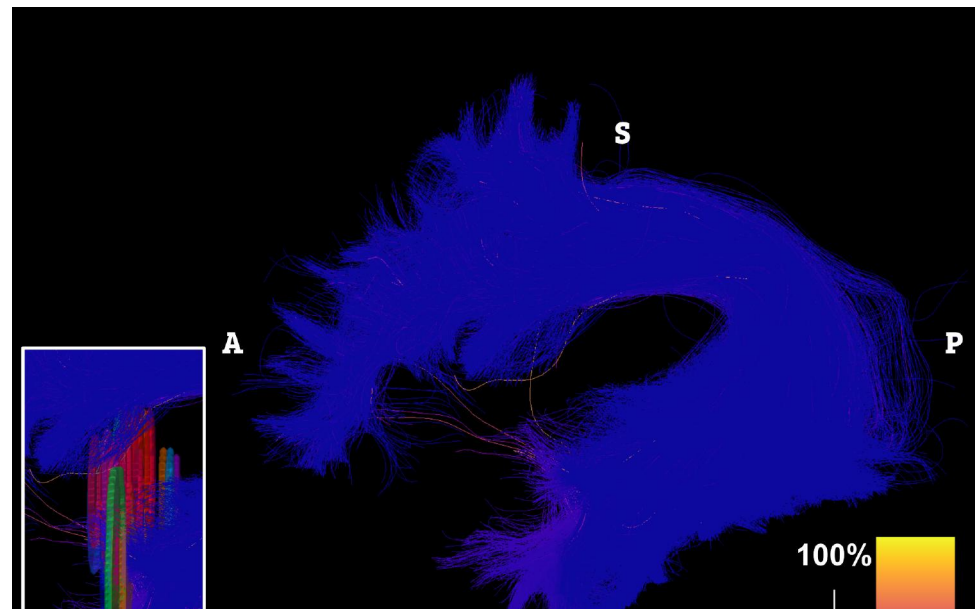
The usage of 2 other ROIs was designed to facilitate rater execution and accelerate the protocol. However, it was observed that the postcentral ROI was highly variable in the A/P axis, sometimes giving the raters a difficult approximate location. This ROI is severely impacted by the variation in the A/P axis. Forcing the streamlines to reach too far back some of the sharp turning streamlines will be excluded, while a ROI position too anteriorly will affect the streamlines terminating at the beginning of the temporal lobe. Most ROI were sufficiently large to correctly encompass the whole desired regions.

Counterintuitively, the best way to improve reproducibility would be to avoid the reference to the postcentral ROI (since it was highly variable in the first place). Identification of the entrance of the temporal lobe could be achieved using a visual description only rather than providing an approximate starting point to the rater. Another, more complex, approach would require a completely new ROI. The high impact of position this ROI (too anteriorly/posteriorly) indicates that locating the stem of the arc (U) of the arcuate fasciculus would be more accurate anatomically and could be more reproducible. This new ROI could be located using the Sylvian fissure to locate the correct axial plane and then identifying a blue hypersignal on the RGB map (when the streamlines switch direction in the curve of the arcuate fasciculus). However, it is difficult to say if nonexperts could reliably find such a region.

Arcuate fasciculus - Temporal stem

This region aims to close off the temporal stem from possible outliers and spurious streamlines. The signal from the uncinate fasciculus could lead streamlines out of the temporal lobe and back into the frontal lobe.

Using the precentral gyrus as a starting point to identify a search region for a strong green hypersignal on the RGB map, a large disk region (20-25mm) had to be drawn. The ROI encompasses the temporal stem while not intersecting the temporal lobe itself or the dorsal pathways



(centrum semiovale), its position is restrained to the insular region.

This region of interest has an exclusion rule and does not affect the final bundles. The spurious streamlines using this passage are extremely rare due to all the rules used to define the arcuate fasciculus. In the current protocol, such streamlines do not exist in sufficient quantity to warrant the use of this “cleaning” ROI.

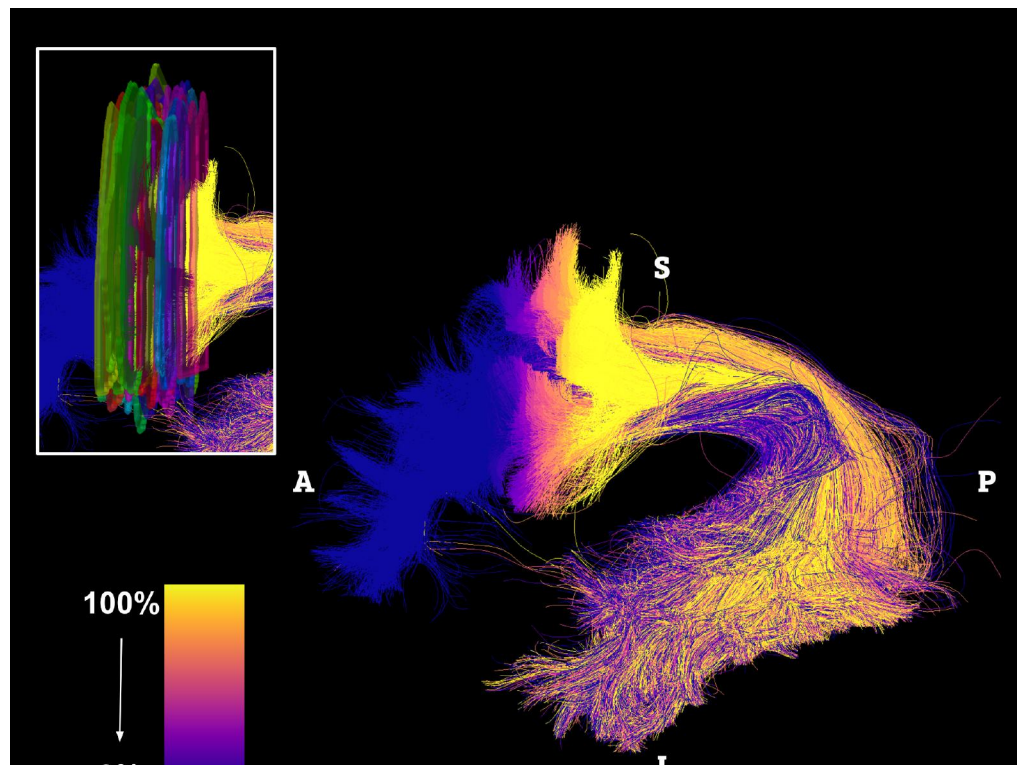
This ROI is a good example of a delineation being a burden on the raters (the variation in size and position indicates it is difficult to locate), not creating any variability and not being useful to the virtual dissection. This ROI should be removed from the protocol as it does not have any positive (or negative) impact, but increases raters' workload.

Arcuate fasciculus - Precentral gyrus

The goal of this region is to force streamlines to reach the frontal lobe to avoid early termination after the centrum semiovale ROI.

The precentral gyrus ROI is the same as the one used for the pyramidal tract. The goal was to reuse as much as possible to decrease the total duration of the protocol.

As opposed to the two other pathways using the precentral gyrus ROI as an exclusion region (low impact), this pathway uses it as an inclusion region (high impact). The large variation in the A/P axis sometimes causes the precentral gyrus to be part of



the final results, but not always. This region leads to the biggest source of spatial and density variability in the protocol (for all bundles). Identification of the precentral gyri planar region is among the hardest ROI of the protocol (also varies from HCP subject to another). This led to the arcuate fasciculus being the pathway with the highest dissection variability.

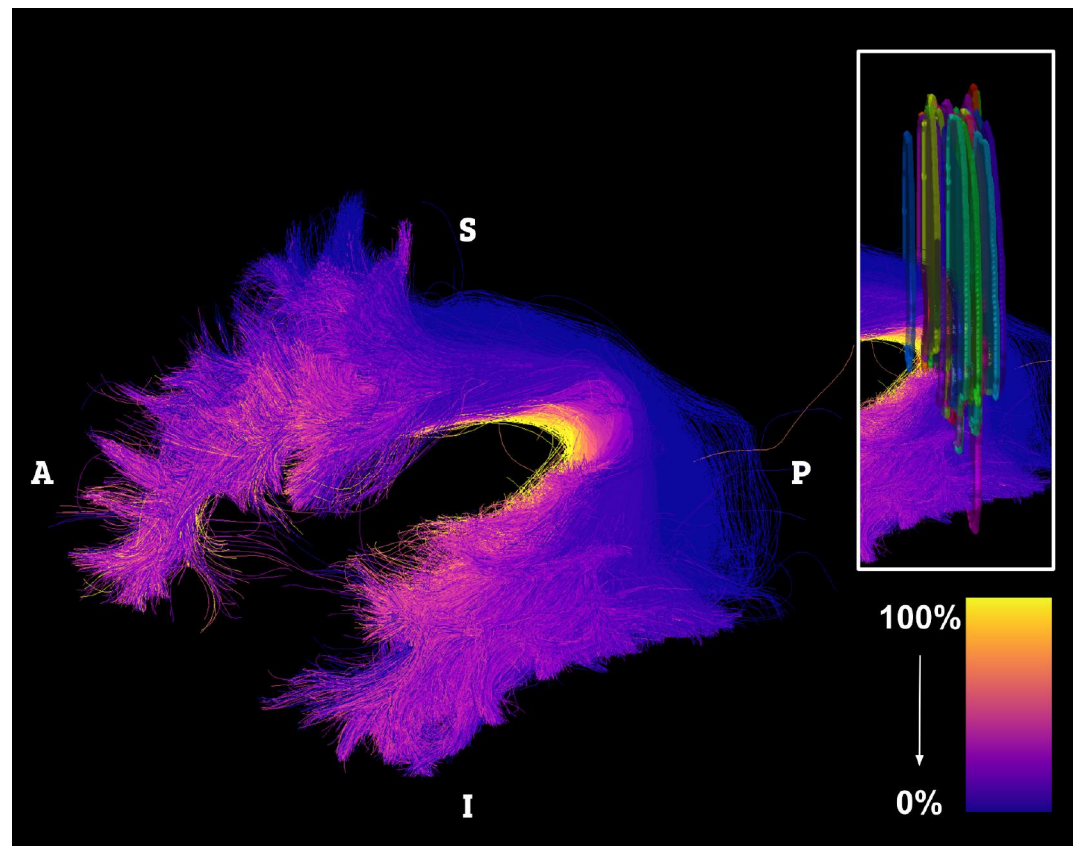
This ROI is a good example of a region of interest needing to be redefined due to anatomical inaccuracies and being a large source of variability. The precentral gyrus planar region was defined as the most anterior point of the gyrus to avoid cutting off the fanning of the corpus callosum and pyramidal tracts (used as exclusion). However, this pathway uses it as an inclusion and so the most posterior point of the precentral gyrus would be needed instead. This leads to two possibilities: 1) Adding a new planar region of interest to the protocol (most posterior portion of the precentral gyrus) or 2) removing this region entirely and moving the centrum semiovale ROI slightly forward to ensure the pathway is effectively reaching into the frontal lobe. Since the identification of landmarks related to the precentral (and postcentral) gyrus seemed to be a difficult task for our raters, we believed option 2) is more appropriate for the protocol. This way the anatomical accuracy is increased from the original protocol, the variability of dissection is decreased and the raters' workload is not increased.

Arcuate fasciculus - Postcentral gyrus

The region aims to further enforce streamlines to curve into the temporal lobe while following the desired path.

The postcentral gyrus ROI is the same as the one used for the pyramidal tract. The goal was to reuse as much as possible to decrease the total duration of the protocol.

The postcentral gyrus ROI has a large variability in the A/P axis, but does not seem to lead to increased variability in the final bundles. Only the ROIs positioned too posteriorly affect a small portion of the inner curve of the arcuate fasciculus. When the region is positioned in a more average position, there is no impact on the spatial extent and only minor differences in density. However, it has a bigger effect on the



superior temporal gyrus and a strong effect close to the parahippocampal gyrus.

With the combination of rules for this pathway virtual dissection, the need for this region is minimal. More explicitly, the temporal entry ROI already forces the path into a curve to reach the temporal lobe. A more careful description of the temporal entry ROI would have a bigger impact on the quality of the final pathway.

Since there is a valid anatomical “reason” for this region/rule, it could be kept as part of the arcuate fasciculus dissection. However, more effort is needed to reduce the variability of the postcentral (and precentral for that matter) gyri. In the current state of the protocol, the postcentral gyrus planar ROI potentially introduces more variability to the arcuate fasciculus dissection than it contributes to its anatomical accuracy.

General recommendations (full-text of section 3.2)

Dissection in standardized space. Virtual dissection protocol should always be performed in a standardized space (e.g. MNI or AC-PC aligned). Either performed before the protocol or in the software itself. The reason for this recommendation is simple: Landmarks do not look the same depending on the brain orientation due to slicing. It is more difficult for raters to locate structures when their shape is affected by the orientation of the slices and when the initial approximate position is harder to find. Relative spatial location is easier to interpret (by human or by computer) in MNI/AC-PC align space (medial to, lateral to). This should be done using linear transformation only (affine or rigid) to avoid deformation of landmarks and the reference images should always be from the subjects themselves (not a template).

Clear instruction on the software does help raters. While the description of regions of interest is crucial, a detailed presentation of the software is likely as important. Presenting functionalities, tips to prevent major pitfalls, avoiding crashes and saving time all are important to reduce the mental workload of a rater. By knowing the software limitations and inconveniences, a virtual dissection protocol can be optimized to reduce the burden on raters and thus increase reproducibility and improve the general experience. Having a single example (video or class) does help raters to use the software more efficiently and reduce critical mistakes. Written text is a reference guide, but step-by-step instruction (or better, a video) is a good way to increase reproducibility.

Virtual dissection workflow. Another way to reduce variability is to establish a workflow for the tasks. Organizing the input, the order in which the data should be loaded in the software, requiring a naming convention for all ROIs and BOIs are all solutions to reduce the workload of raters (and likely reduce the stress associated with the tasks). Saving all ROIs and BOIs for the raters to

acknowledge that each ROIs and BOIs is accounted for is a simple, yet effective verification method. This also allows for a quick verification at the end of the protocols (does every dataset have X ROIs and Y BOIs, do they have the right filename, is the file size consistent enough or are there empty files).

Getting in the zone. While the accuracy of a virtual dissection does not significantly change over time, each rater workflow (software manipulation, files structure, details about the protocol, filename convention) needs some time to get up to speed. It is good practice to set time to focus and perform a few virtual dissections at the time and to always complete a dataset before stopping. This not only avoids mistakes, but also reduces the overall time needed to complete large datasets. It was observed that each protocol 'restart' was taking around 5-10 minutes when the actual process of virtual segmentation of a single subject was 20-30 minutes.

Being too strict is worse than being permissive. Using very small ROIs or using too many exclusion regions is not only harder for raters, but it leads to poor results. Using permissive and easier to locate ROIs leads to good results. Combining multiple ROIs will facilitate the process of virtual dissection. A good balance of inclusion and exclusion is a better approach than relying mostly on one or the other, but using too many ROIs can increase the workload of raters.

Using other ROIs as landmarks. This is generally a good practice to facilitate and accelerate the virtual dissection process. By providing a lot of details on specific ROIs, the subsequent regions should become easier to locate as they reference previously drawn ones. However, this is true only if used for ROIs that are highly consistent. A region with high variability will 'transfer' its variability to any other region referencing it (e.g. central centrum semiovale being based on the center position between the precentral and postcentral gyri: Too variable). Any ROIs being used as reference should receive special attention to its description and limit as much as possible its intrinsic variability.

Generating more than one ROI from the same landmarks. Only re-use ROIs if they have the same impact, if not a new ROI may be needed. For example, if a ROI is used twice, once as inclusion and once as exclusion, and they lead to different effects on their respective dissection, it could be more reproducible and easier for raters to create two separate ROIs from the same set of instructions. Once a landmark is located, one copy could be created with a small translation and the other with a bigger radius, for example. This would allow that time to be used efficiently and optimal ROIs are drawn for each use in a different set of rules.

Providing ranges. Each ROIs should have an approximate initial location and an approximate size to accommodate the anatomical variability of subjects. When in a standardized space, a range of slices can be provided to facilitate location. The exact value should be avoided, raters should be instructed to start from an approximate location and scroll until something specific is located.

Locating landmarks. It appears that instructions using the RGB map are easier to follow than the ones using only T1. This is likely because RGB contains hypersignal of 3 main colors, much easier to describe and locate than a general area of 'gray' on the T1. It also appears that descriptions that involve multiple orientations at once are difficult (axial, coronal sagittal). Landmarks that are perpendicular to the slicing orientation should be prioritized to limit switching from one plane to the other (in the software).

Optimal ROIs design. When using the **inclusion rule** for a stem/core, the positioning along the pathway is less important, the focus should be targeted at the perpendicular positioning (perpendicular to a strongly oriented core). A larger diameter is likely better than a smaller one. A small diameter is prone to large effects if badly positioned and this can lead to losing a large number of streamlines. However, larger diameters can lead to spurious streamlines, but this is often solved by using a combination of ROIs and is, therefore, less problematic. When using the **exclusion rule**, bigger is often better as long as you do not accidentally cut in half the pathway somewhere else (e.g. middle sagittal plane to prevent commissural streamlines accidentally blocking the brainstem). For exclusion rules, adding a few millimeters in the opposite direction of what is supposed to be stopped provides a good margin of error to reduce variability. Indeed, an exclusion region placed too close to the landmark (as some raters will likely *cut/exclude* before the landmark and ruin the dissection) is much worse than one being positioned slightly after the landmark.

Increasing complexity. The virtual dissection protocol should start with basic descriptions of the bundles of interests, a summary of the steps that will follow and a general overview of all regions that will be drawn. Afterward, each ROI should start with a description of how it will be used and overall meaning to inform the raters. Then, providing an approximate starting position (using major, easy-to-find structures) is important to reduce uncertainty for raters. The next step is to provide details on how to locate a specific landmark from this previously identified approximate location (e.g. scroll slowly until you locate X, you should not go past Y). Additionally, ROIs' instruction should start with a fast (and less accurate) set of instructions to delineate the region followed by a slower (more accurate) detailed instruction. This way a protocol is not only more intuitive to follow and provides redundant verification of validity to the raters, but the same documents/course can be used for different applications. Finally, adding a verification step so the raters can confirm if they made a mistake or not (e.g. top portion of the ROI should not go above W or go below Z) limits doubtfulness.

Cleaning ROIs. Some protocols attempt to increase the reproducibility of bundles by allowing cleaning ROIs to manually remove spurious streamlines. This operation can inadvertently lead to more variability by leaving too much room for interpretation and potential software manipulation mistakes. If cleaning ROIs are needed (not anatomically meaningful, but simply to clean outliers from tractography limitations), they should be described and justified, follow the previously listed recommendations and all cleaning ROIs should be accounted for. It should not be left to the raters to decide how big, how many, where cleaning ROIs are needed.